

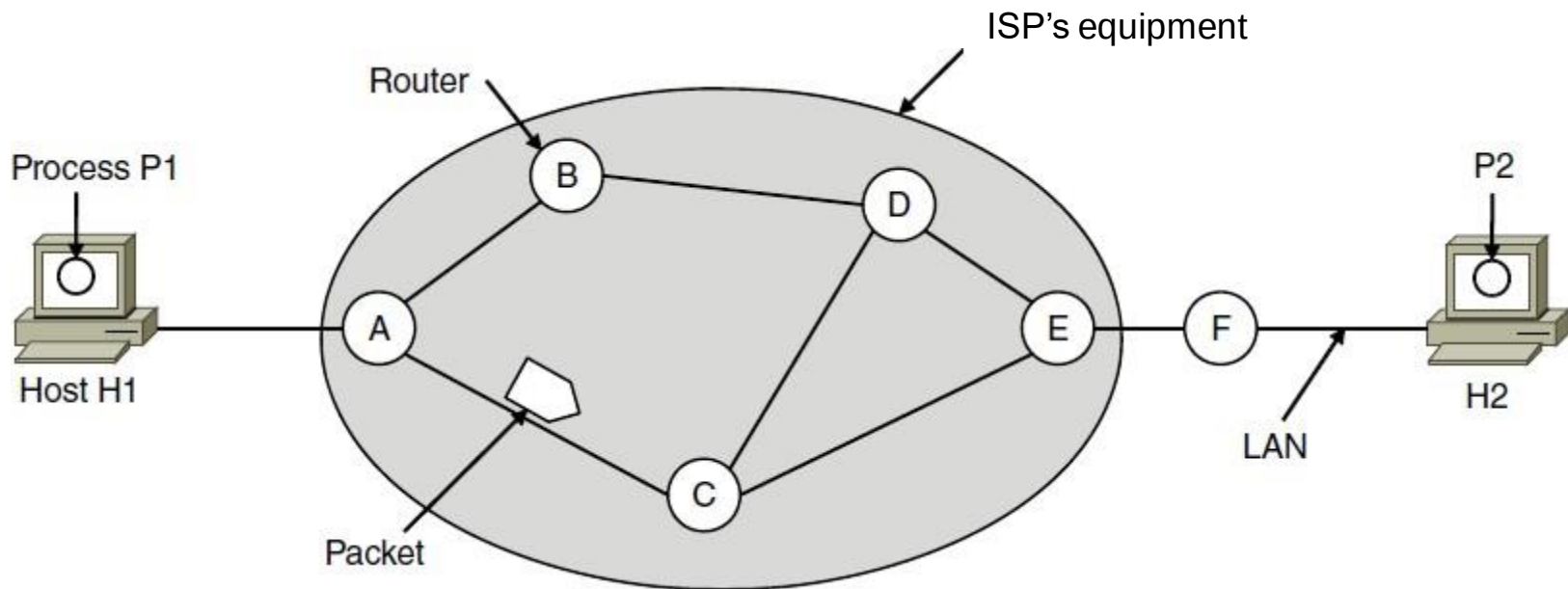
The Network Layer

Chapter 5

Network Layer Design Issues

- Store-and-forward packet switching
- Services provided to transport layer
- Implementation of connectionless service
- Implementation of connection-oriented service
- Comparison of virtual-circuit and datagram networks

Store-and-Forward Packet Switching

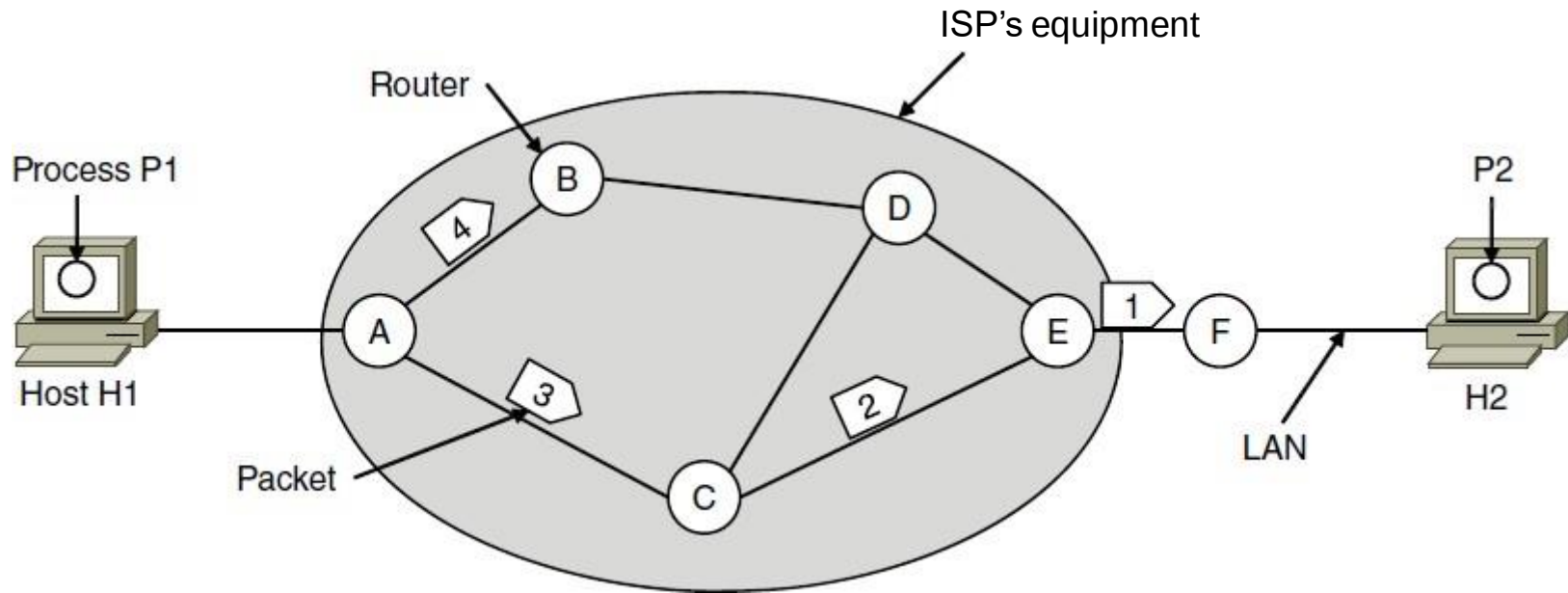


The environment of the network layer protocols.

Services Provided to the Transport Layer

- Services independent of router technology.
- Transport layer shielded from number, type, topology of routers.
- Network addresses available to transport layer use uniform numbering plan even across LANs and WANs

Implementation of Connectionless Service



A's table (initially)

A	
B	B
C	C
D	B
E	C
F	C

Dest. Line

A's table (later)

A	
B	B
C	C
D	B
E	D
F	D

C's Table

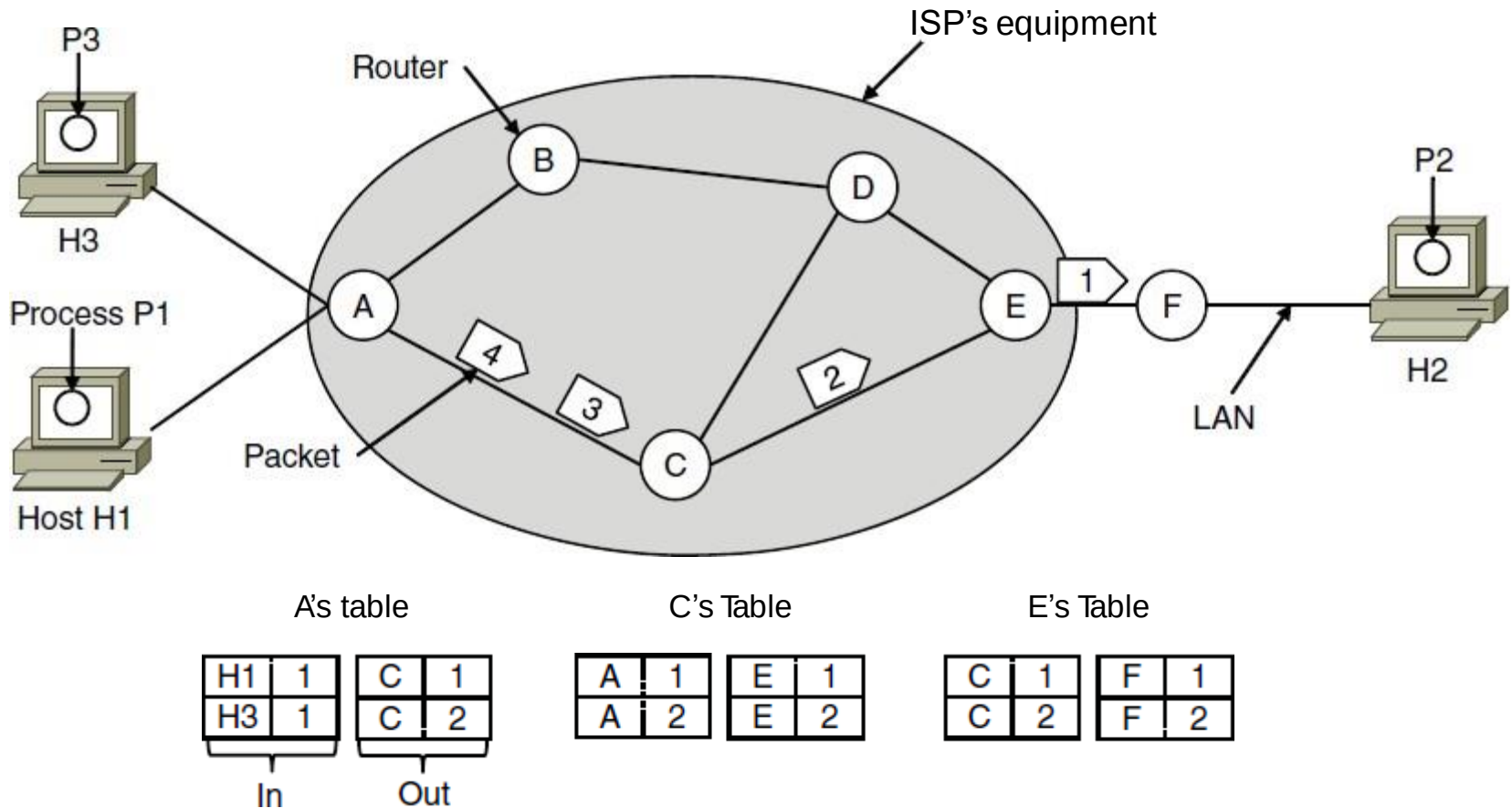
A	A
B	A
C	
D	E
E	E
F	E

E's Table

A	C
B	D
C	C
D	D
E	
F	F

Routing within a datagram network

Implementation of Connection-Oriented Service



Routing within a virtual-circuit network

Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Comparison of datagram and virtual-circuit networks

Routing Algorithms (1)

- Optimality principle
- Shortest path algorithm
- Flooding
- Distance vector routing
- Link state routing
- Routing in ad hoc networks

Routing Algorithms (2)

- Broadcast routing
- Multicast routing
- Anycast routing
- Routing for mobile hosts
- Routing in ad hoc networks

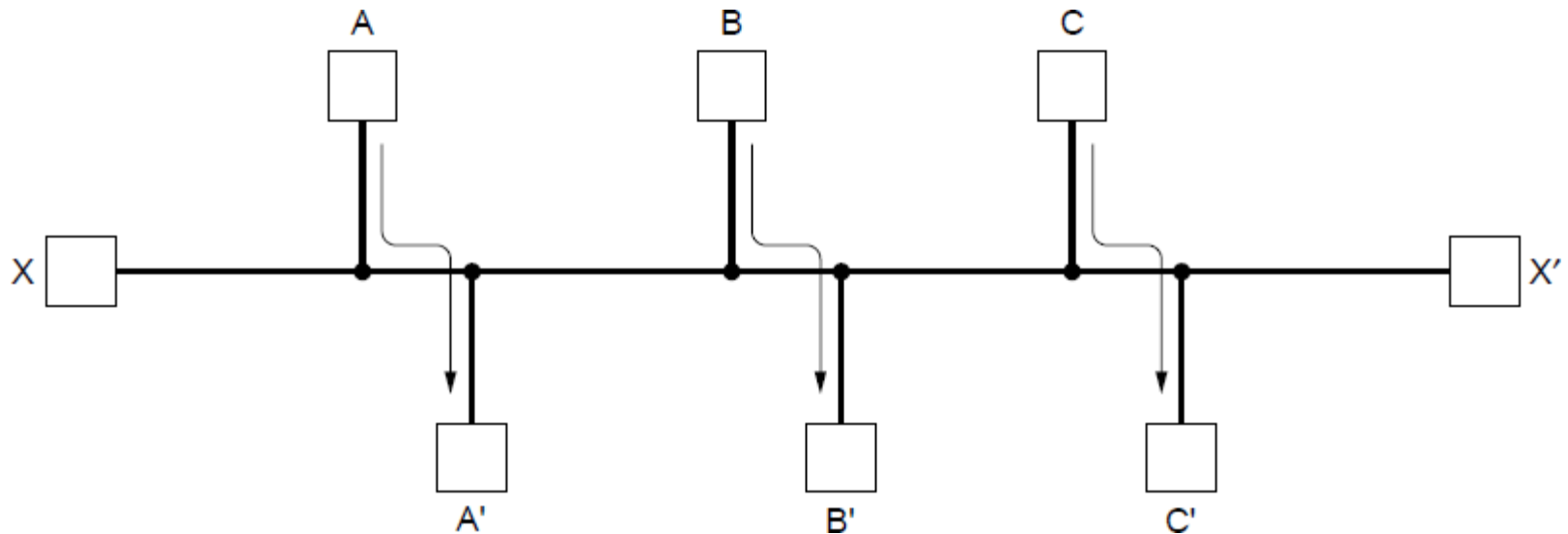
Routing Vs Forwarding

- Routing is making the decision which routes to use, and forwarding, is what happens when a packet arrives
- Router has two processes inside it. One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing tables. This process is **forwarding**. The other process is responsible for filling in and updating the routing tables.

Routing Algorithms

- Desirable properties of routing algorithm:
 - Correctness – Right path is chosen
 - Simplicity – Ease of use and implementation
 - Robustness – reliability in event of failure
 - Stability - converge to fixed set of paths
 - Fairness – Fair to all hosts
 - Efficiency – Utilization, delay

Fairness vs. Efficiency



Network with a conflict between fairness and efficiency.

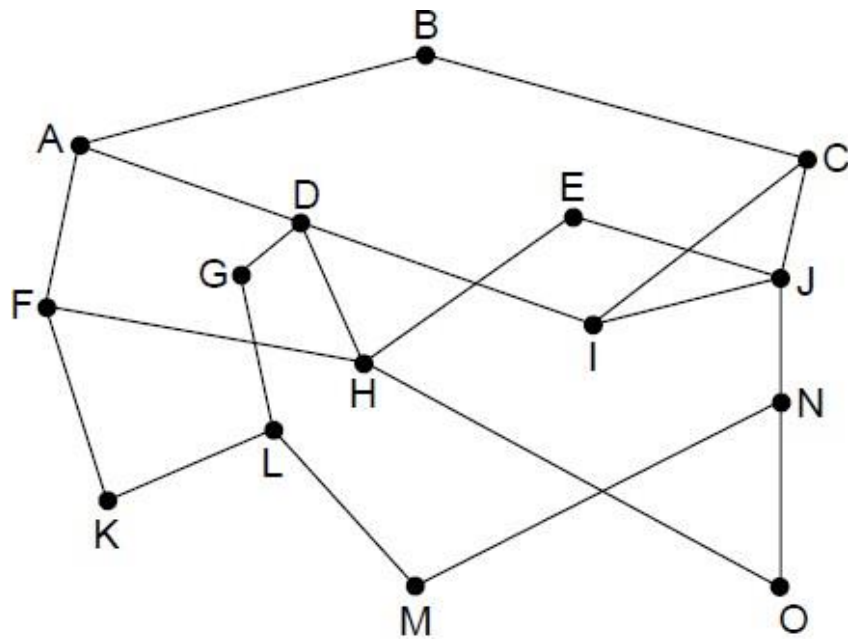
Routing Algorithms

- Types of routing algorithm:
 - Nonadaptive RA – do not base their routing decisions on any measurements or estimates of the current topology and traffic. Instead the routes are computed in advance, offline and downloaded to routers when network boots. Eg Static Routing
 - Adaptive RA - change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well. They differ in where they get their information, when they change the routes, and what metric is used for optimization. Eg Distance Vector Routing, Link State Routing

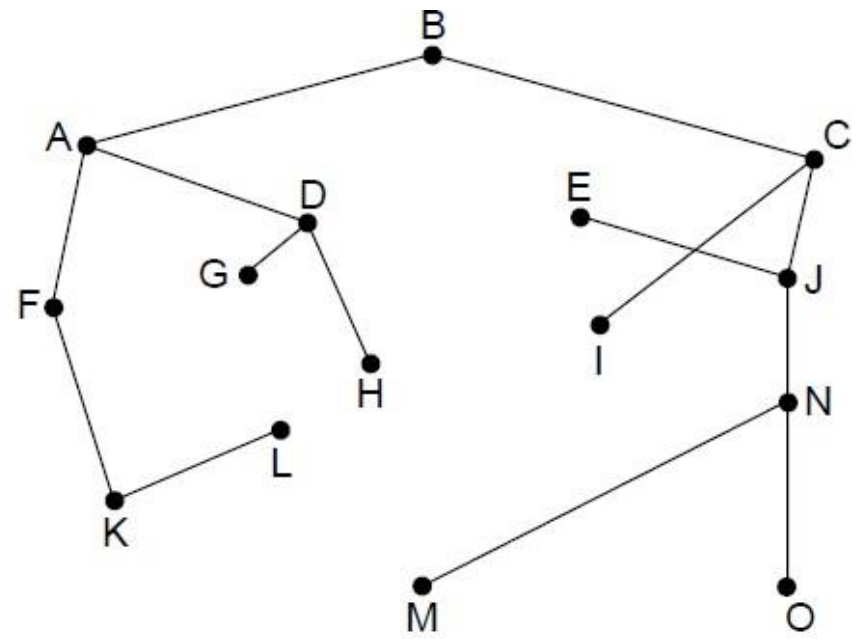
The Optimality Principle

Statement about optimal routes without regard to network topology or traffic

It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route.



(a)



(b)

(a) A network. **(b)** A sink tree for router B.

Flooding

A simple method to send a packet to all network nodes

Each node floods a new packet received on an incoming link by sending it out all of the other links

Flooding generates vast numbers of duplicate packets, in fact, an infinite number unless some measures are taken to damp the process.

One such measure is to have a hop counter contained in the header of each packet that is decremented at each hop.

The packet being discarded when the counter reaches zero.

Ideally, the hop counter should be initialized to the length of the path from source to destination.

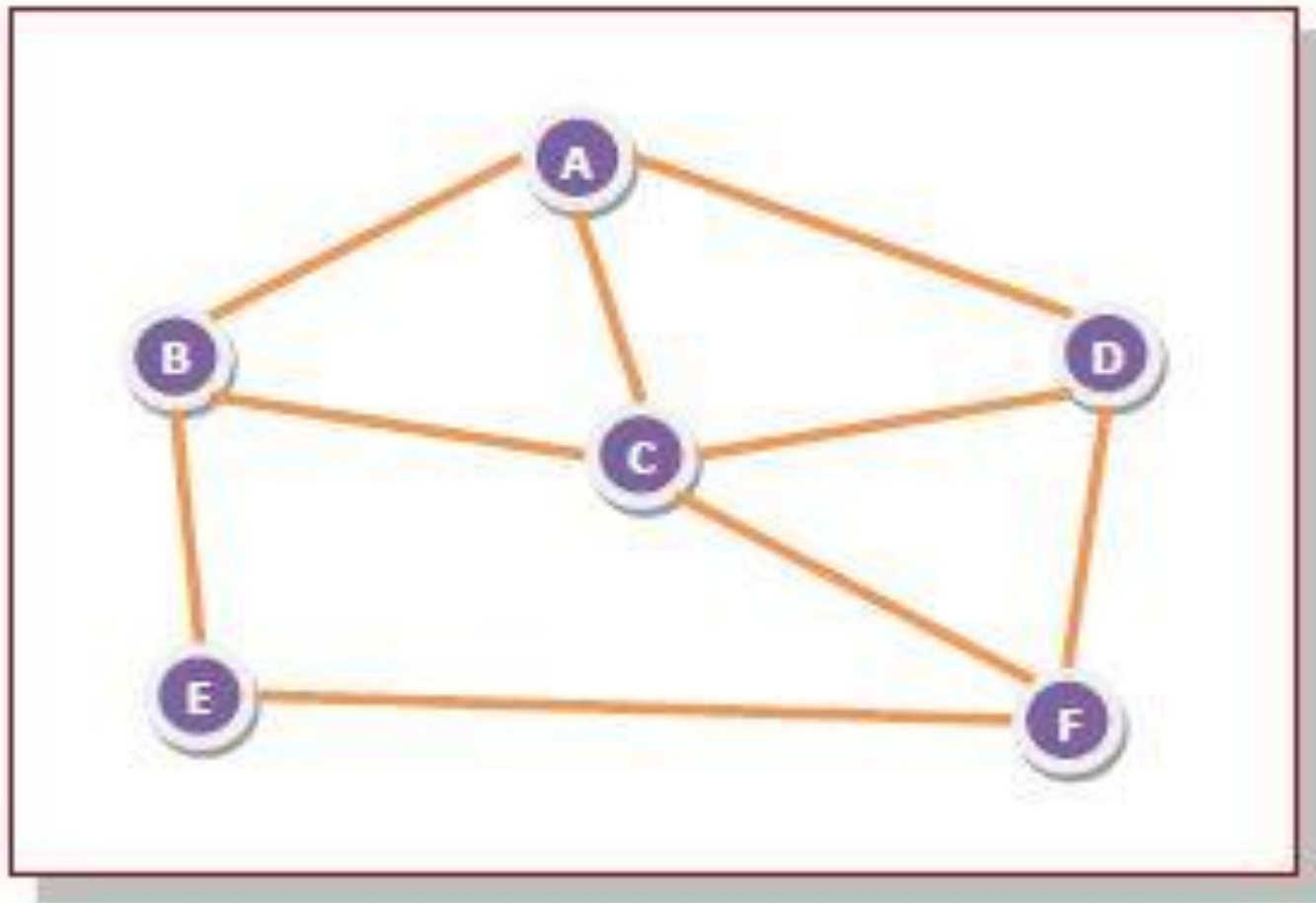
If the sender does not know how long the path is, it can initialize the counter to the worst case i.e. the full diameter of the network.

First, it ensures that a packet is delivered to every node in the network.

This may be wasteful if there is a single destination that needs the packet, but it is effective for broadcasting information.

Flooding will find a path if one exists, to get a packet to its destination(e.g. in a military network located in a war zone).

Example



Example

An incoming packet to A, will be sent to B, C and D.

B will send the packet to C and E.

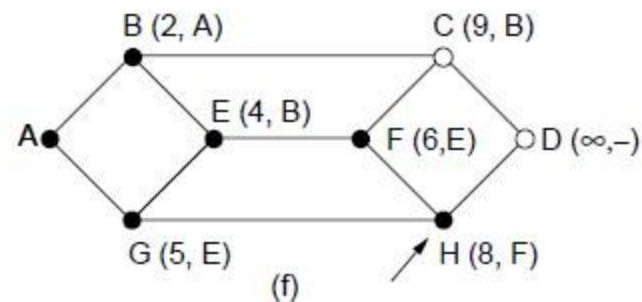
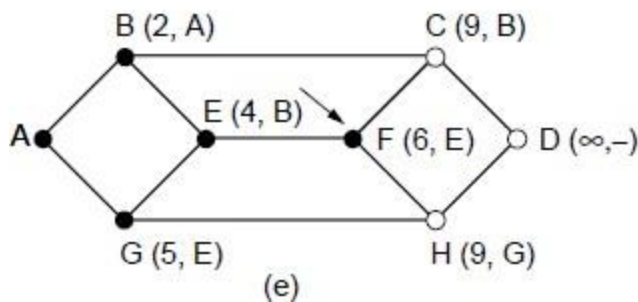
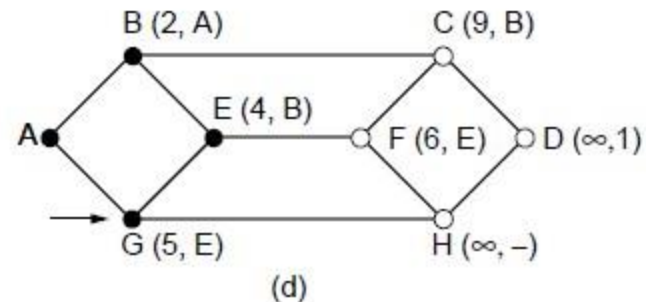
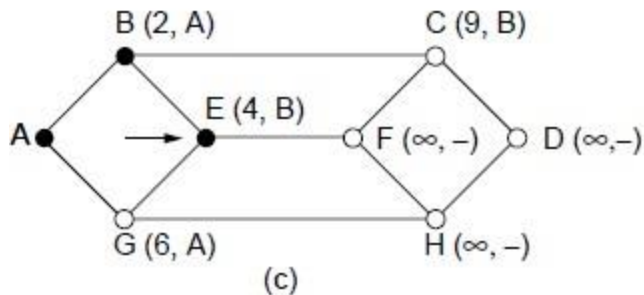
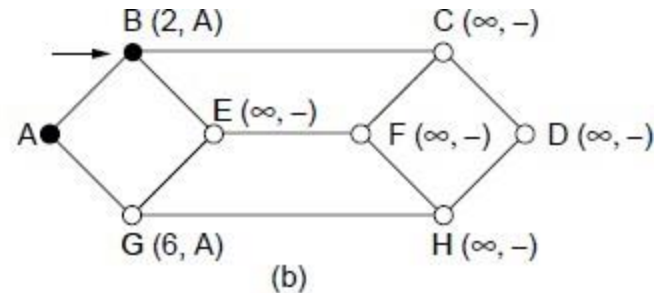
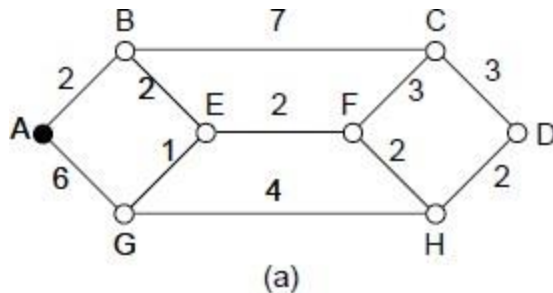
C will send the packet to B,D and F.

D will send the packet to C and F.

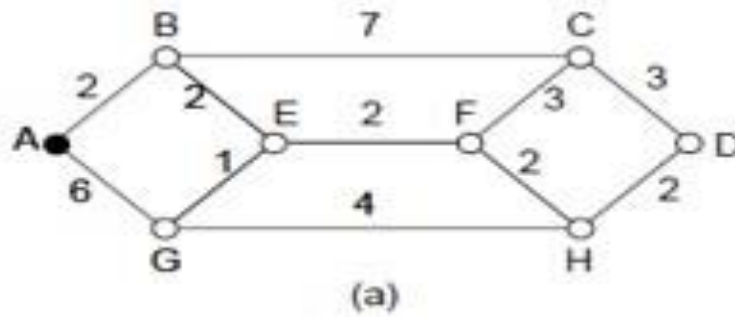
E will send the packet to F.

F will send the packet to C and E.

Shortest Path Algorithm (1)

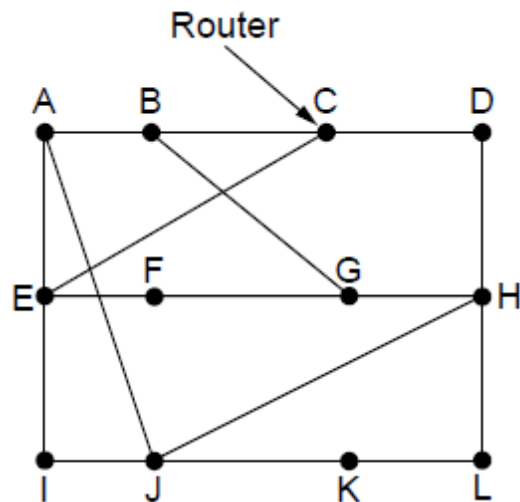


The first five steps used in computing the shortest path from *A* to *D*. The arrows indicate the working node



A	B	C	D	E	F	G	H
	2 A	∞	∞	∞	∞	6 A	∞
+2 B	✓	9 B	∞	4 B	∞	6 A	∞
+4 E	✓	9 B	∞	✓	6 A	5 E	∞
+5 G	✓	9 B	∞	✓	6 A	✓	9 G
+6 F	✓	9 B	∞	✓	✓	✓	8 F
+8 H	✓	9 B	10 H	✓	✓	✓	✓
+9 C	✓	✓	10 H	✓	✓	✓	✓

Distance Vector Routing



(a)

To	A	I	H	K	New estimated delay from J ↓ Line	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6
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Vectors received from J's four neighbors

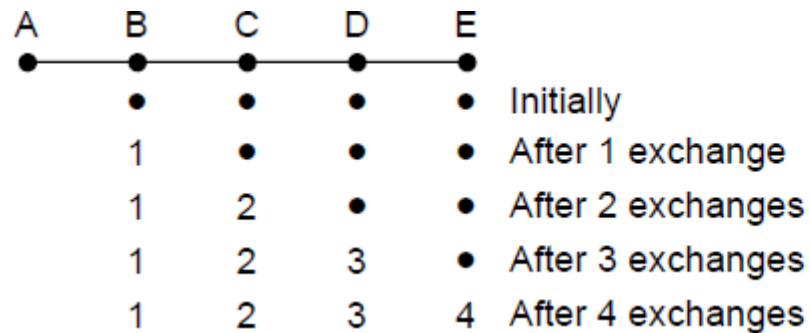
New routing table for J	
8	A
20	A
28	I
20	H
17	I
30	I
18	H
12	H
10	I
0	—
6	K
15	K

(b)

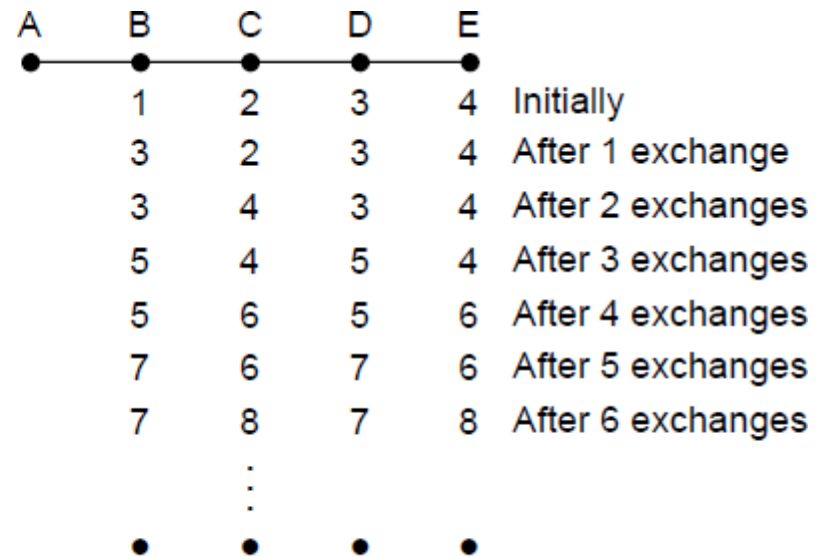
(a) A network.

(b) Input from A, I, H, K, and the new routing table for J.

The Count-to-Infinity Problem



(a)



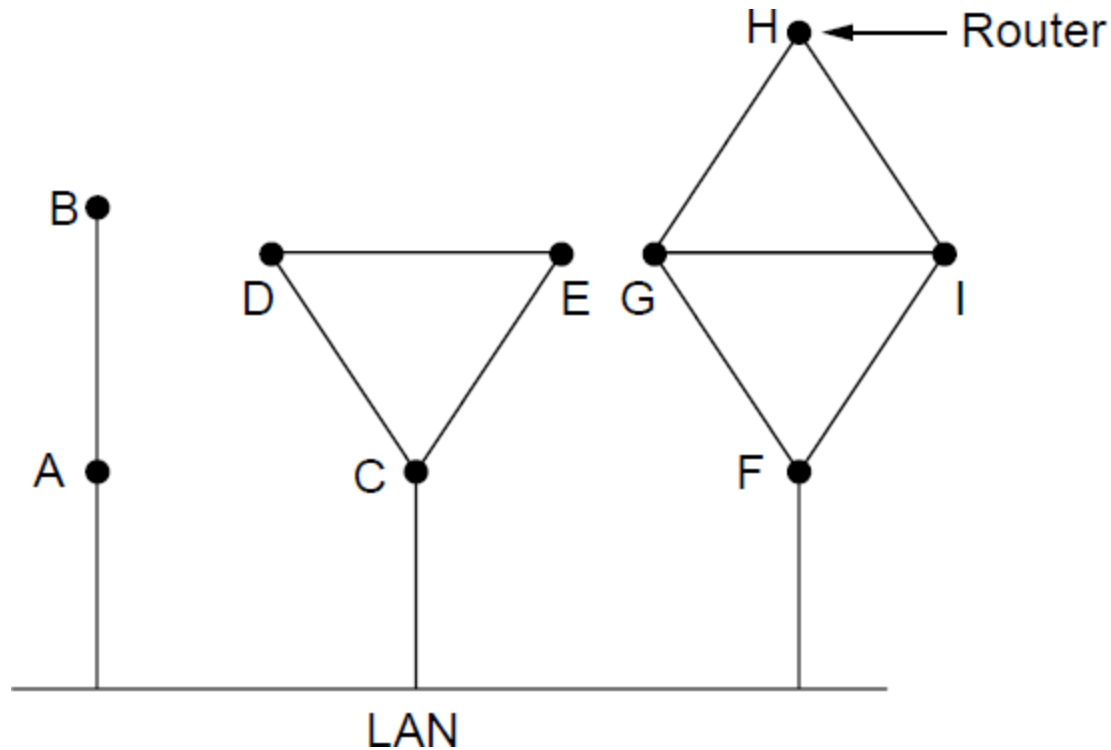
(b)

The count-to-infinity problem

Link State Routing

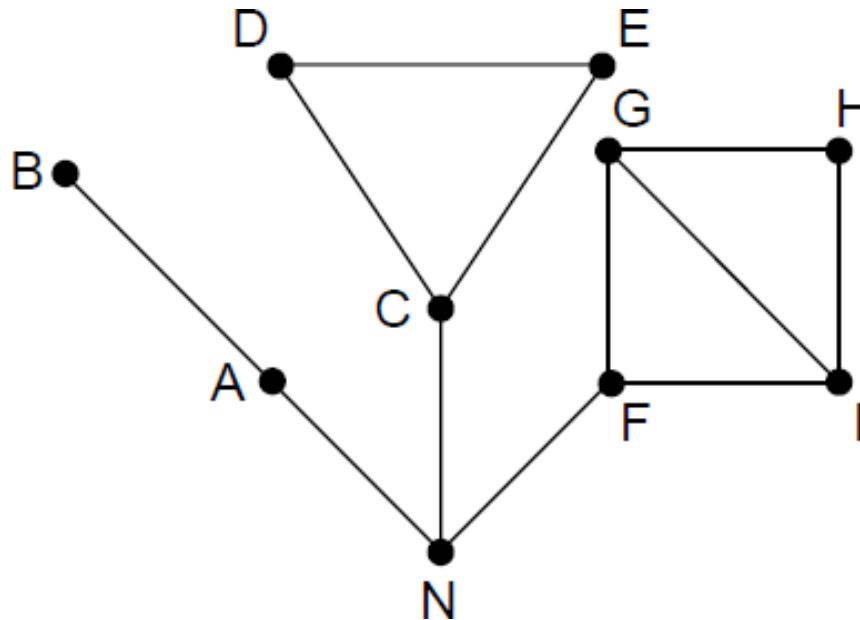
- Discover neighbors, learn network addresses.
- Set distance/cost metric to each neighbor.
- Construct packet telling all learned.
- Send packet to, receive packets from other routers.
- Compute shortest path to every other router.

Learning about the Neighbors (1)



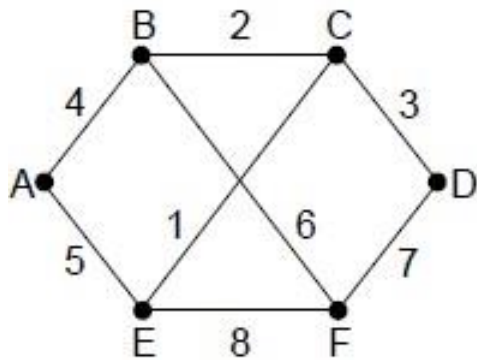
Nine routers and a broadcast LAN.

Learning about the Neighbors (2)



A graph model of previous slide.

Building Link State Packets



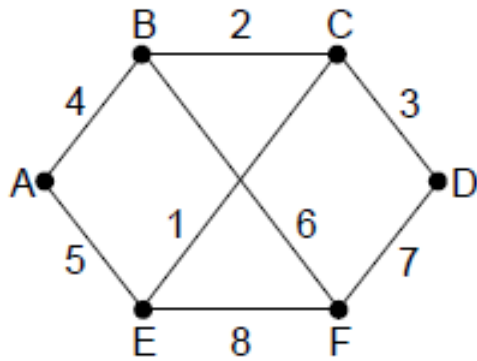
(a)

		Link		State		Packets	
A		B		C		D	
Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age	
B	4	A	4	B	2	C	3
E	5	C	2	D	3	F	7
		F	6	E	1		

(b)

(a) A network. (b) The link state packets for this network.

Distributing the Link State Packets



(a)

Link		State		Packets	
A		B		C	
Seq.		Seq.		Seq.	
Age		Age		Age	
B	4	A	4	B	2
E	5	C	2	D	3
		F	6	E	1

(b)

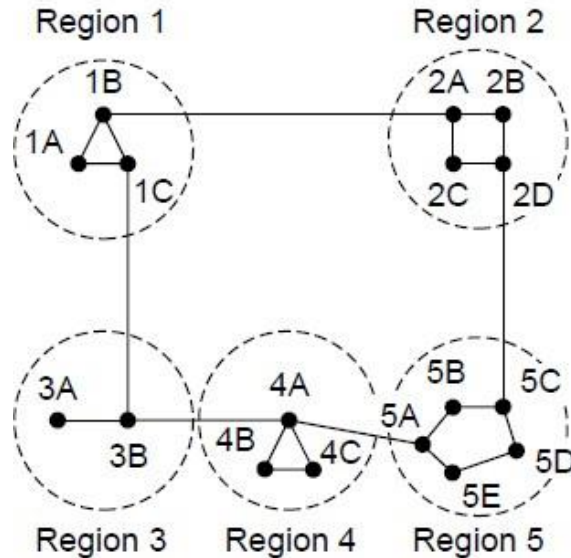
Source	Seq.	Age	A C F			A C F			Data
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

The packet buffer for router *B* in previous slide

Hierarchical Routing

- As networks grow in size, the routing tables grow proportionally
- Not only is router memory consumed by ever-increasing tables, but more CPU time is needed to scan them and more bandwidth is needed to send status reports about them.
- The routers are divided into what we will call **regions**.
- Two levels may not be sufficient for huge networks
- It may be necessary to group the regions into clusters, the clusters into zones, the zones into groups
- How many levels should the hierarchy have?
- For example, consider a network with 720 routers. If there is no hierarchy, each router needs 720 routing table entries. If the network is partitioned into 24 regions of 30 routers each, each router needs 30 local entries plus 23 remote entries for a total of 53 entries. If a three-level hierarchy is chosen, with 8 clusters each containing 9 regions of 10 routers, each router needs 10 entries for local routers, 8 entries for routing to other regions within its own cluster, and 7 entries for distant clusters, for a total of 25 entries.

Hierarchical Routing



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

Hierarchical routing.

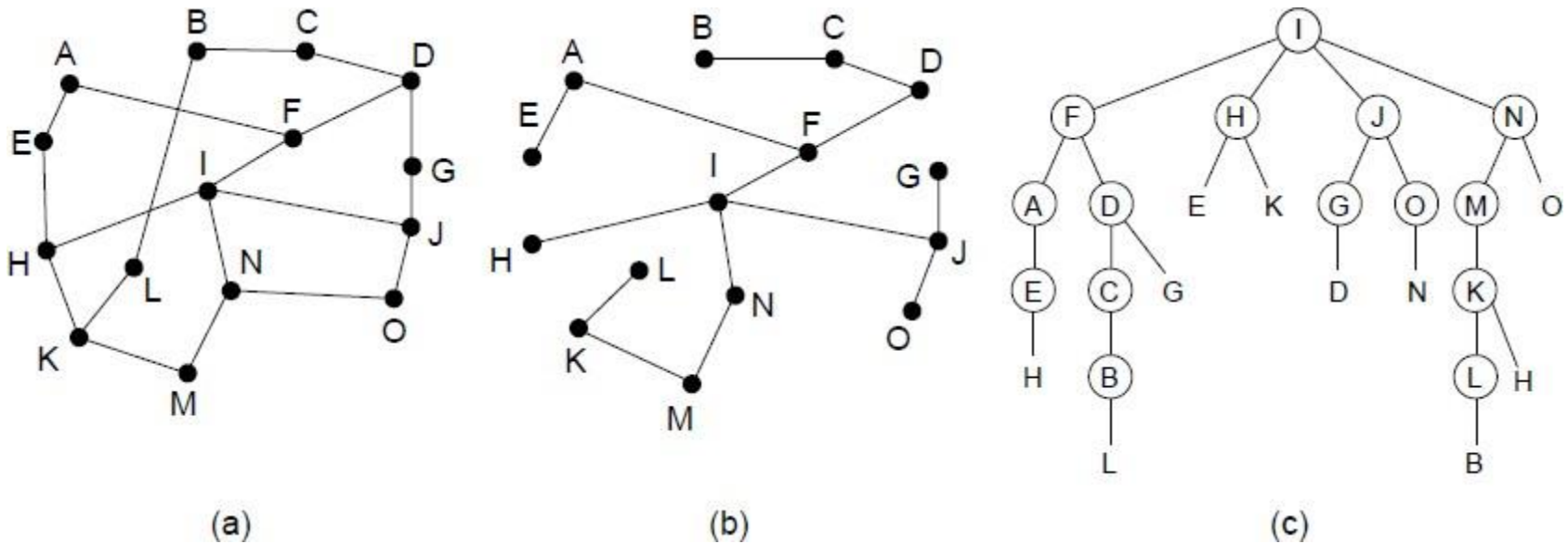
Broadcast Routing

- Sending a packet to all destinations simultaneously is called **broadcasting**
- One broadcasting method that requires no special features from the network is for the source to simply send a distinct packet to each destination.
- An improvement is **multidestination routing**, in which each packet contains either a list of destinations or a bit map indicating the desired destinations.
- When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed. The router generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line
- A better broadcast routing technique is **flooding**

Broadcast Routing – Reverse Path Forwarding

- When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets *toward* the source of the broadcast.
- If so, there is an excellent chance that the broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router.
- This being the case, the router forwards copies of it onto all links except the one it arrived on.
- If, however, the broadcast packet arrived on a link other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.

Broadcast Routing

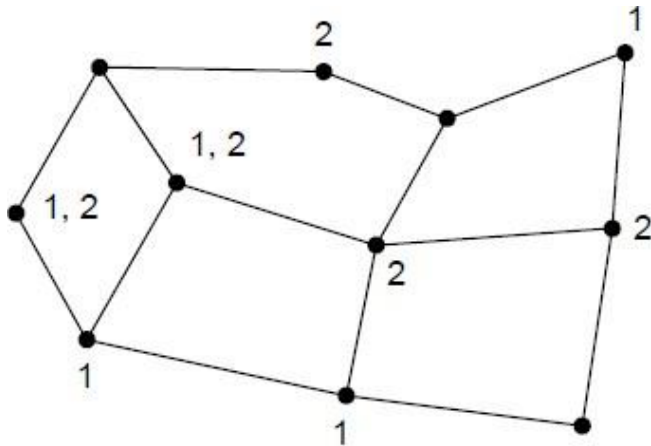


Reverse path forwarding. (a) A network. (b) A sink tree for I.
(c) The tree built by reverse path forwarding.

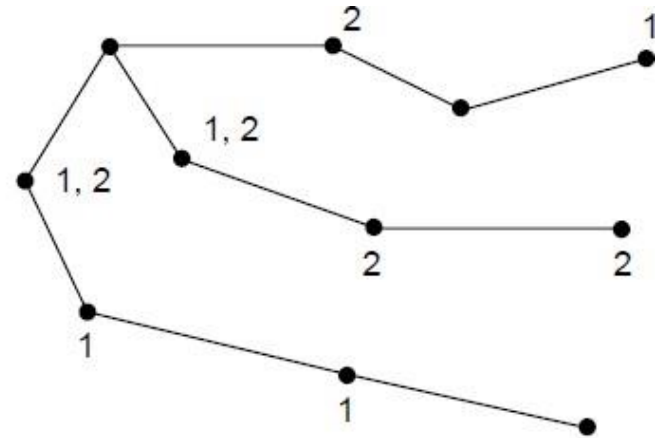
Multicast Routing

- Multicasting is a special case of broadcasting
- All multicasting schemes require some way to create and destroy groups and to identify which routers are members of a group
- Assume that each group is identified by a multicast address and that routers know the groups to which they belong
- If the group is dense, broadcast is a good start but it reaches all routers. The solution is pruning the broadcast tree by removing links that do not lead to members
- If OSPF is used, each routers knows complete topology, they can construct its own pruned spanning tree for each sender to the group. Eg. **MOSPF (Multicast OSPF)**

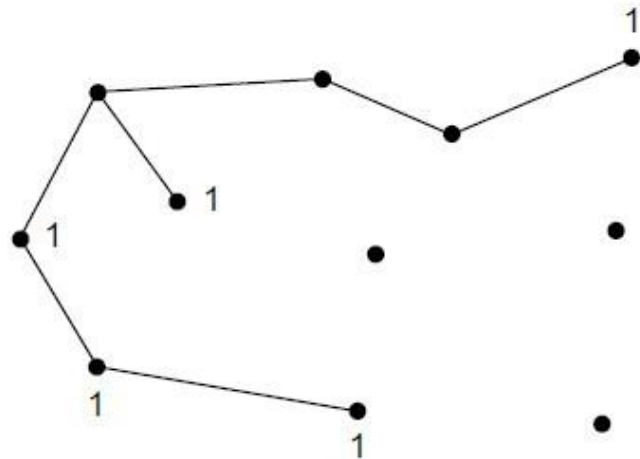
Multicast Routing (1)



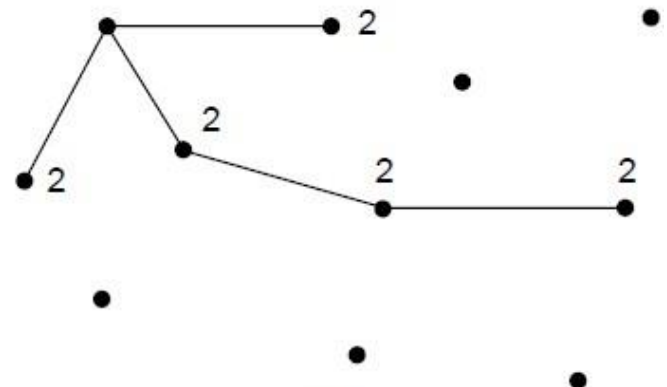
(a)



(b)



(c)

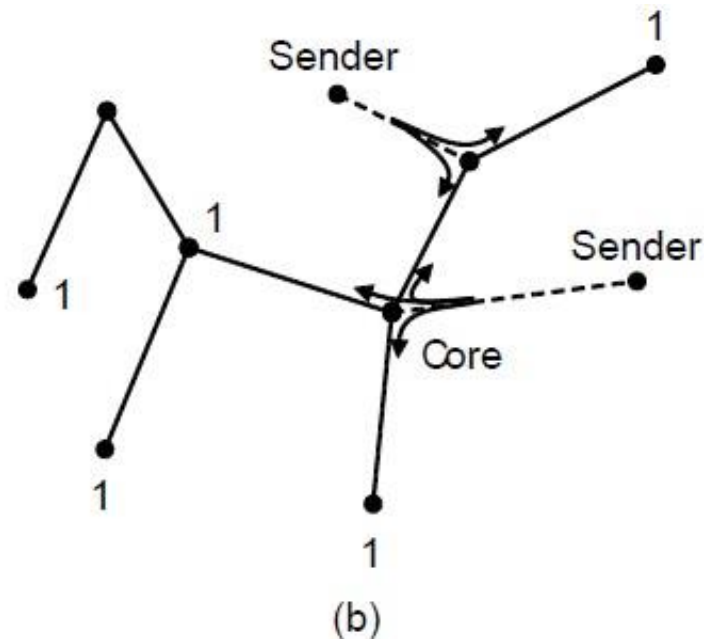
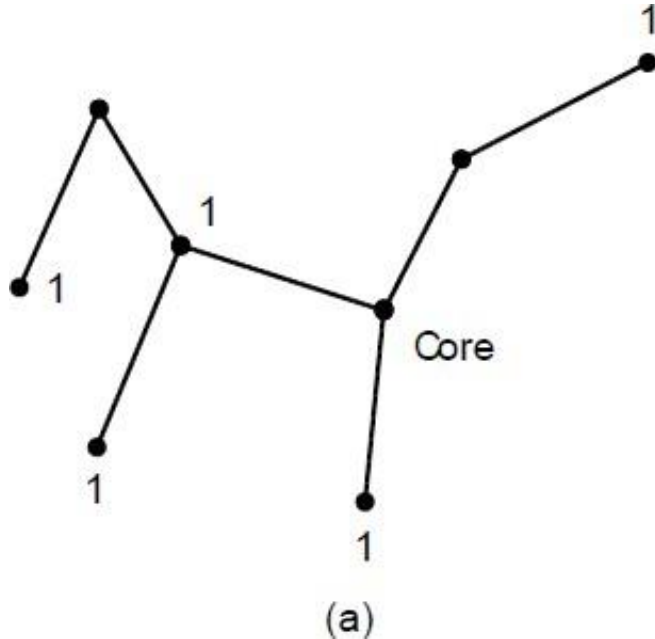


(d)

(a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

Multicast Routing (2)

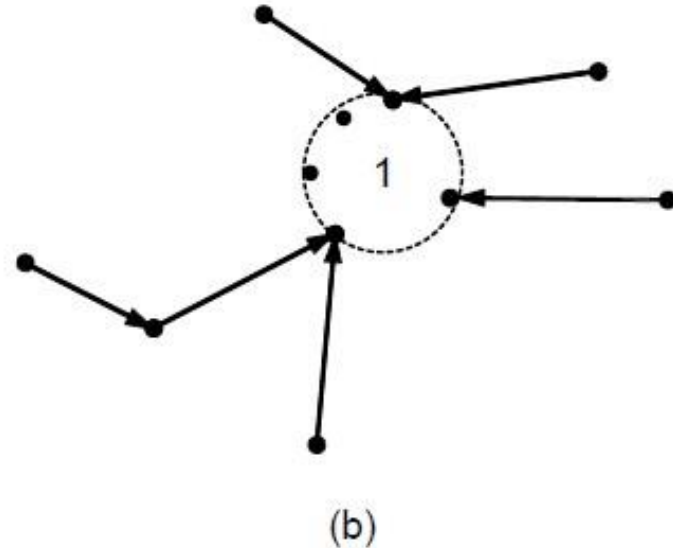
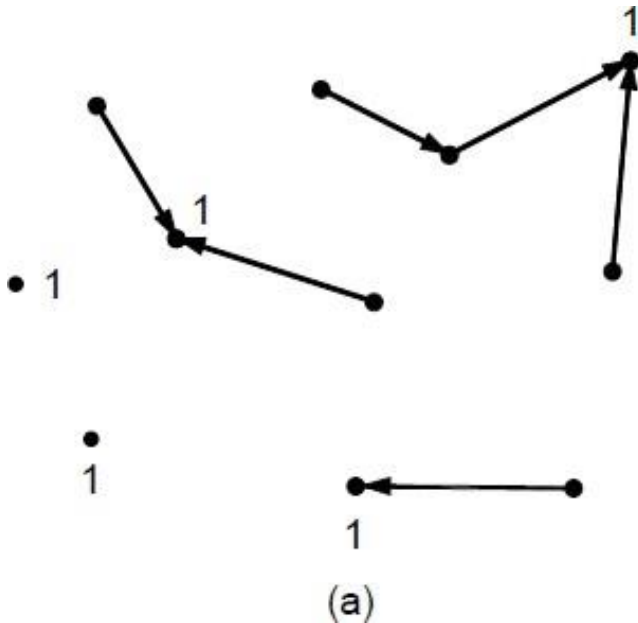
- All of the routers agree on a root (called the **core** or **rendezvous point**)
- Build the tree by sending a packet from each member to the root
- The tree is the union of the paths traced by these packets
- To send to this group, a sender sends a packet to the core



- a) Core-based tree for group 1.
- b) Sending to group 1.

Anycast Routing

- A packet is delivered to the nearest member of a group
- Can be achieved using regular DV or LSR
- All members of the group will be given same address, say 1
- This procedure works because the routing protocol does not realize that there are multiple instances of destination 1. That is, it believes that all the instances of node 1 are the same node



a) Anycast routes to group 1.

b) Topology seen by the routing protocol.

Congestion Control Algorithms (1)

- Approaches to congestion control
- Traffic-aware routing
- Admission control
- Traffic throttling
- Load shedding

Congestion Control

Too many packets present in the network causes packet delay and loss that degrades performance. This situation is called **congestion**.

The network and transport layers share the responsibility for handling congestion.

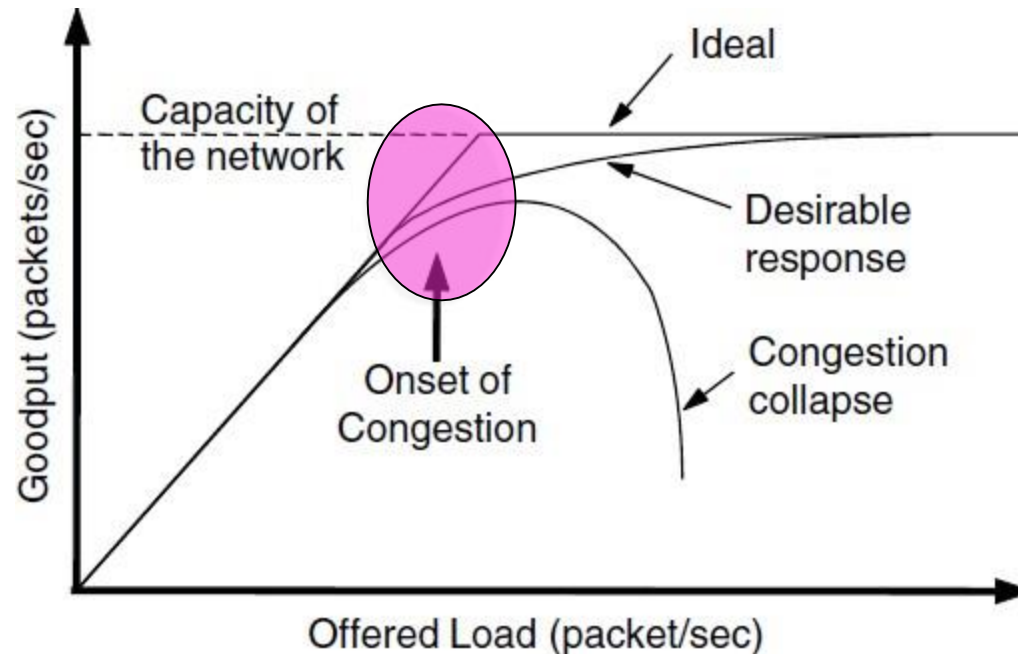
Since congestion occurs within the network, it is the network layer that directly experiences it and must ultimately determine what to do with the excess packets.

However, the most effective way to control congestion is to reduce the load that the transport layer is placing on the network. This requires the network and transport layers to work together.

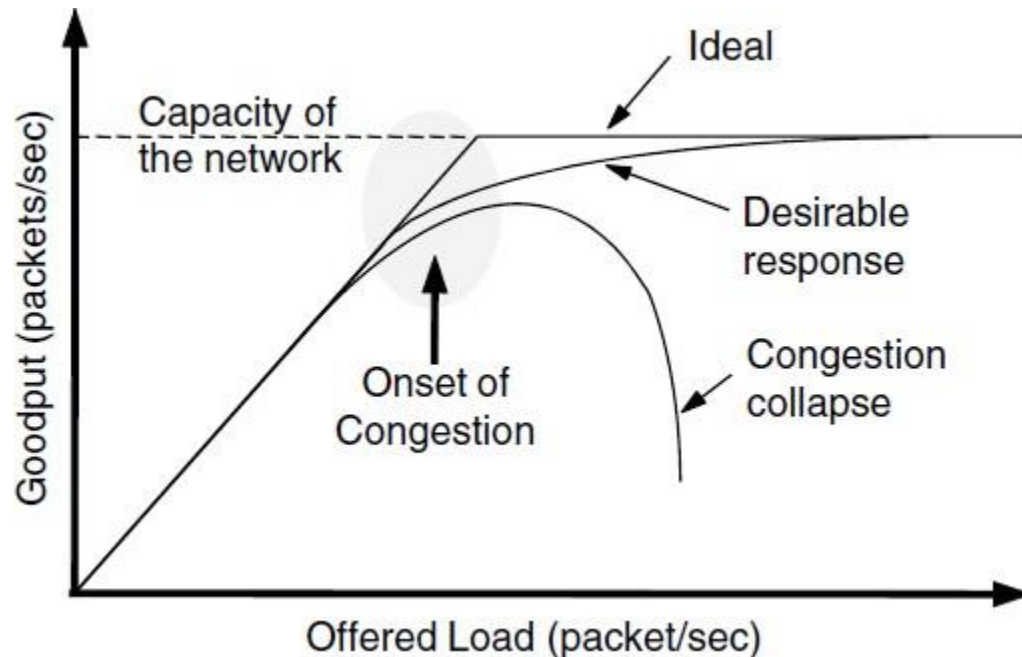
Congestion Control (2)

Congestion results when too much traffic is offered;
performance degrades due to loss/retransmissions

- Goodput (useful packets) trails offered load



Congestion Control Algorithms (2)



When too much traffic is offered, congestion sets in and performance degrades sharply.

Congestion Control (1)

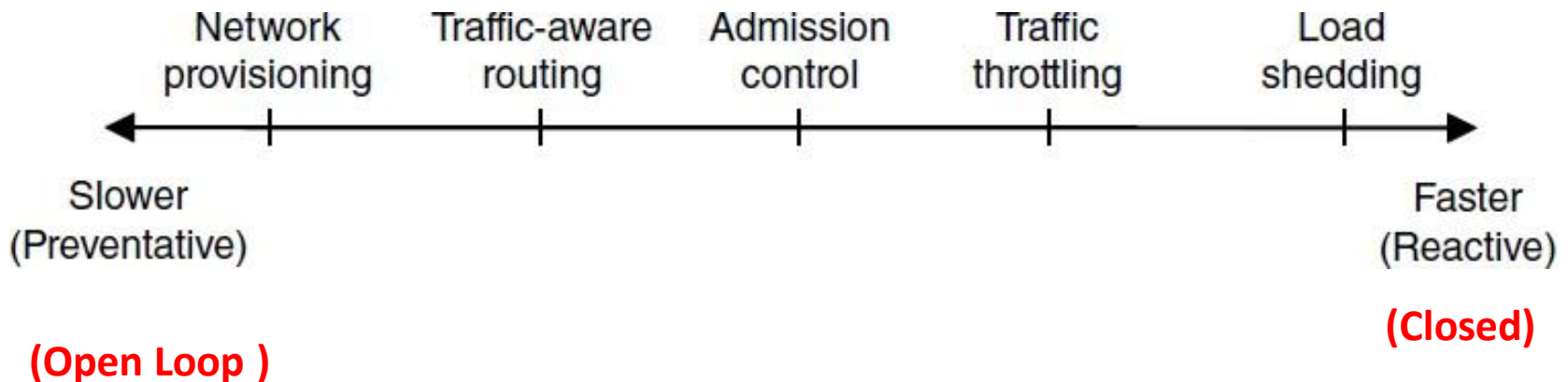
Handling congestion is the responsibility of the Network and Transport layers working together

- We look at the Network portion here
 - Network provisioning »
 - Traffic-aware routing »
 - Admission control »
 - Traffic throttling »
 - Load shedding »

Congestion Control (3) – Approaches

Network must do its best with the offered load

- Different approaches at different timescales
- Nodes should also reduce offered load (Transport)



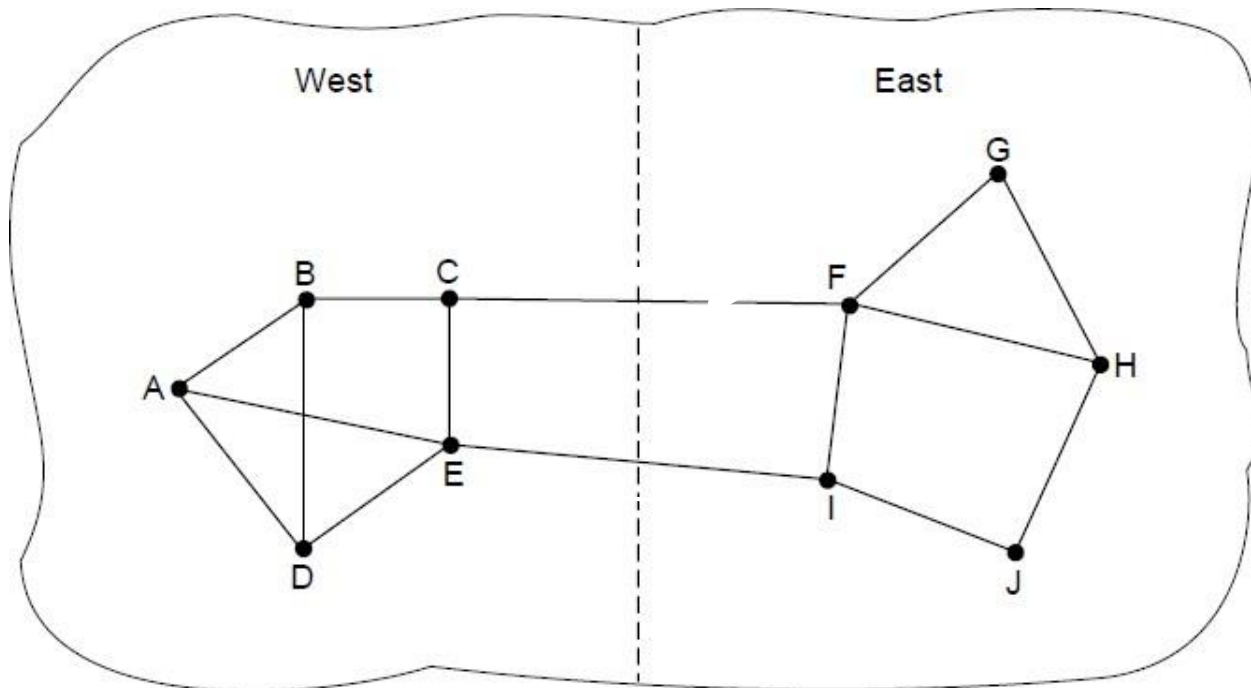
Traffic Aware Routing

- The most direct way to do this is to set the link weight to be a function of the (fixed) link bandwidth and propagation delay plus the (variable) measured load or average queuing delay.
- Consider the network in the next slide, which is divided into two parts East and West connected by two links CF and EI.
- The routing tables may oscillate wildly leading to erratic routing and many other potential problems.
- If load is ignored and only bandwidth and propagation delay are considered, this problem does not occur.
- Two techniques can contribute to the successful solution. The first is **multipath routing**, in which there can be multiple paths from a source to a destination.
- The second one is for the **routing scheme to shift the traffic across routes slowly** enough that is able to converge.

Traffic-Aware Routing

Choose routes depending on traffic, not just topology

- E.g., use *EI* for West-to-East traffic if *CF* is loaded

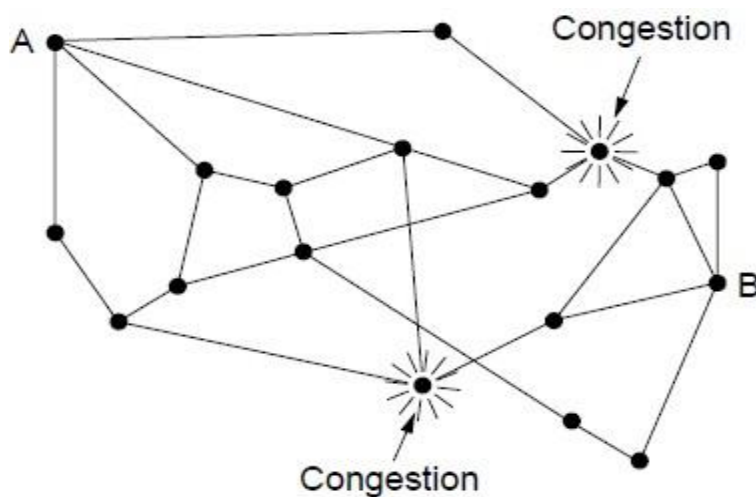


Admission Control

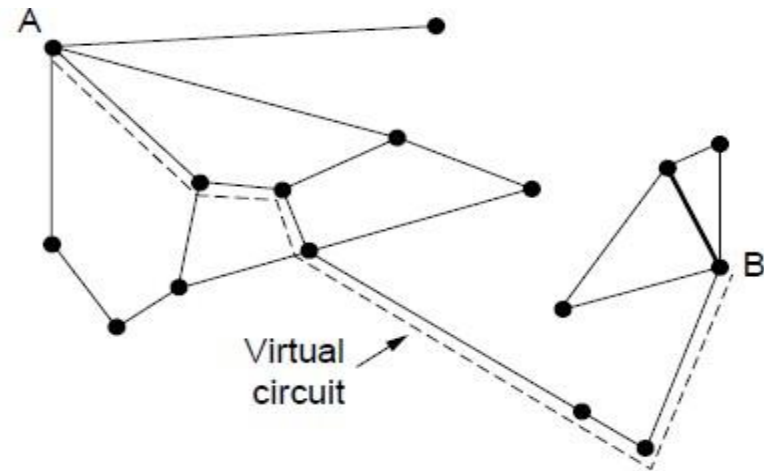
- The idea: do not set up a new virtual circuit unless the network can carry the added traffic without being congested.
- Traffic is often described in terms of its **rate** and **shape**.
- A commonly used descriptor that captures this effect is the **leaky bucket** or **token bucket**.
- Armed with traffic descriptions, the network can decide whether to admit the new virtual circuit.
- The traffic description is the **service agreement** for what the network will guarantee its users.
- Admission control can also be combined with traffic aware routing by considering routes around traffic hotspots as part of the setup procedure.

Admission Control

- a) Admission control allows a new traffic load only if the network has sufficient capacity.
- b) new connections can be refused if they would cause the network to become
- c) congested.



Network with some congested nodes



Uncongested portion and route AB around congestion

CN5E by Tanenbaum &

Wetheral © Pearson

Traffic Throttling

- Senders adjust their transmissions to send as much traffic as the network can readily deliver.
- The network aims to operate just before the onset of congestion.
- When congestion is imminent, it must tell the senders to throttle back their transmissions and slow down.
- It is used for **congestion avoidance**.
- Routers must determine when congestion is approaching, ideally before it has arrived. To do so each router can continuously monitor the resource it is using-
- three possibilities are: the utilization of the output links, the buffering of queued packets inside the router and the number of packets that are lost due to insufficient buffering.

$$d_{\text{new}} = \alpha d_{\text{old}} + (1 - \alpha)s$$

This is called Exponential Weighted Moving Average(EWMA) d is occupancy, α is the weight in $[0,1]$ and s is the instantaneous value

Traffic Throttling

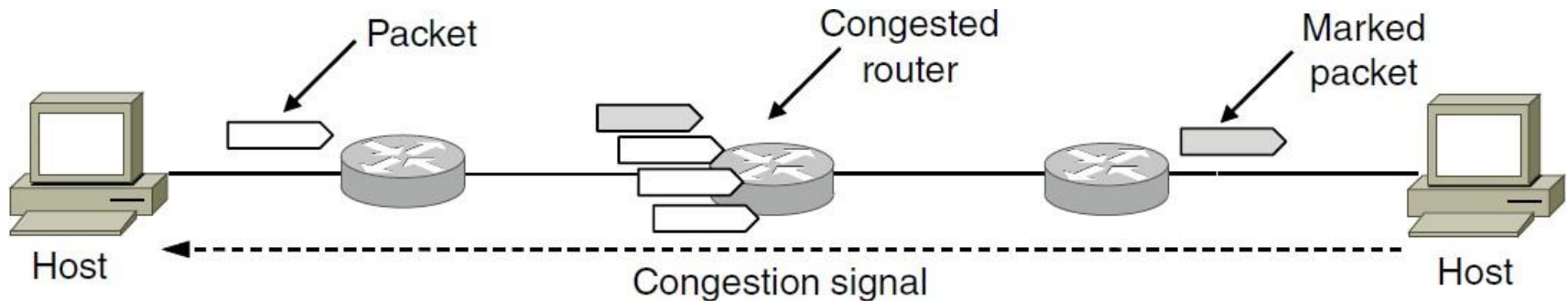
When the network delivers the packet, the destination can note that there is congestion and inform the sender when it sends a reply packet.

The sender can then throttle its transmissions as before.

Congestion notification may be either Implicit or Explicit.

ECN (Explicit Congestion Notification) marks packets and receiver returns signal to sender

ECN may be in either forward or backward direction.



Choke Packet Technique

Choke packet scheme is mechanism where each link is monitored to examine how much utilization is taking place.

If the utilization goes beyond a certain threshold limit, the link goes to a warning and a special packet, called choke packet is sent to the source.

On receiving the choke packet, the source reduced the traffic in order to avoid congestion.

Choke Packet Technique

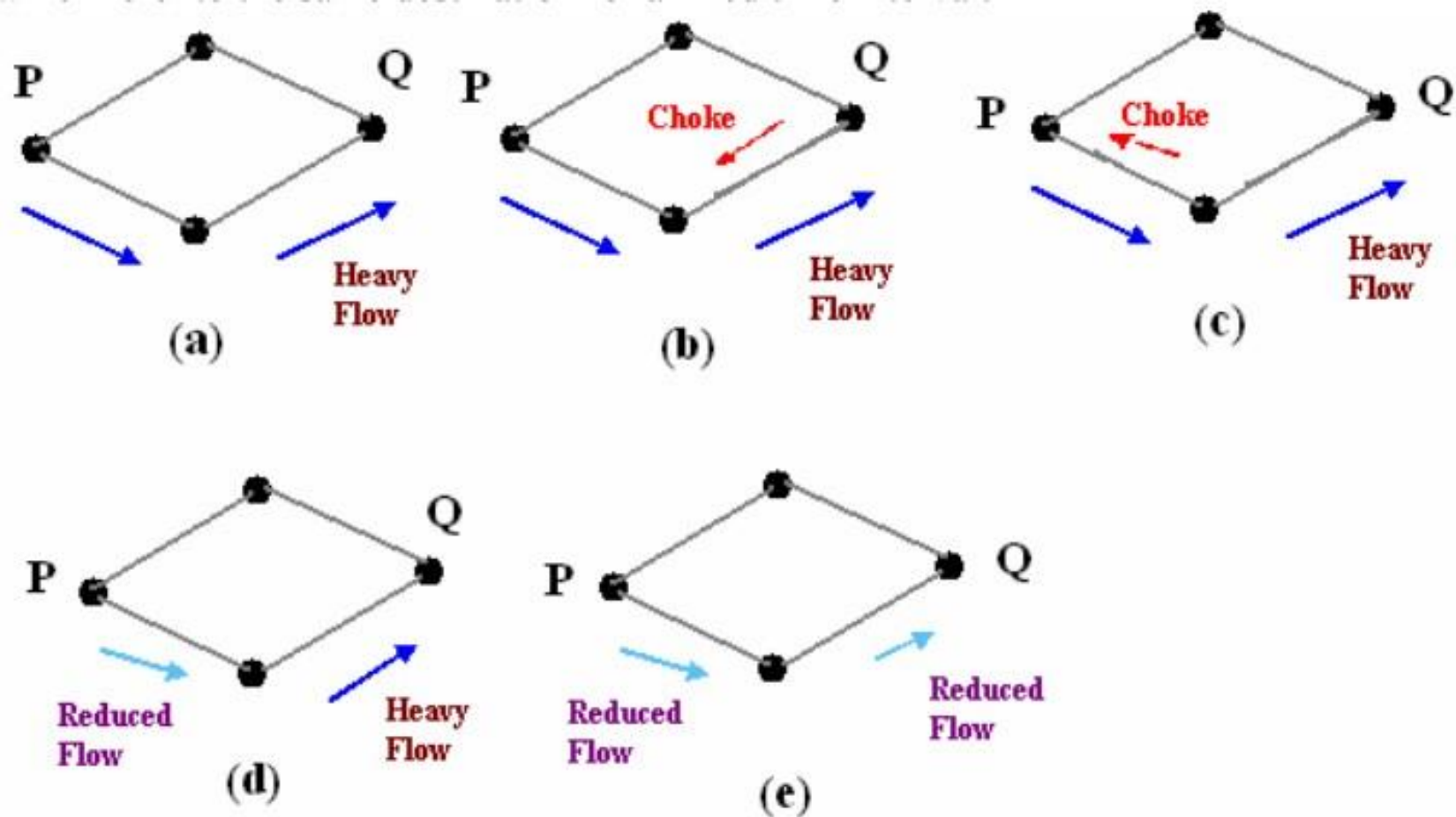


Figure 7.5.6 Depicts the functioning of choke packets, (a) Heavy traffic between nodes P and Q, (b) Node Q sends the Choke packet to P, (c) Choke packet reaches P, (d) P reduces the flow and send a reduced flow out, (e) Reduced flow reaches node Q

Hop-by Hop Choke Packets

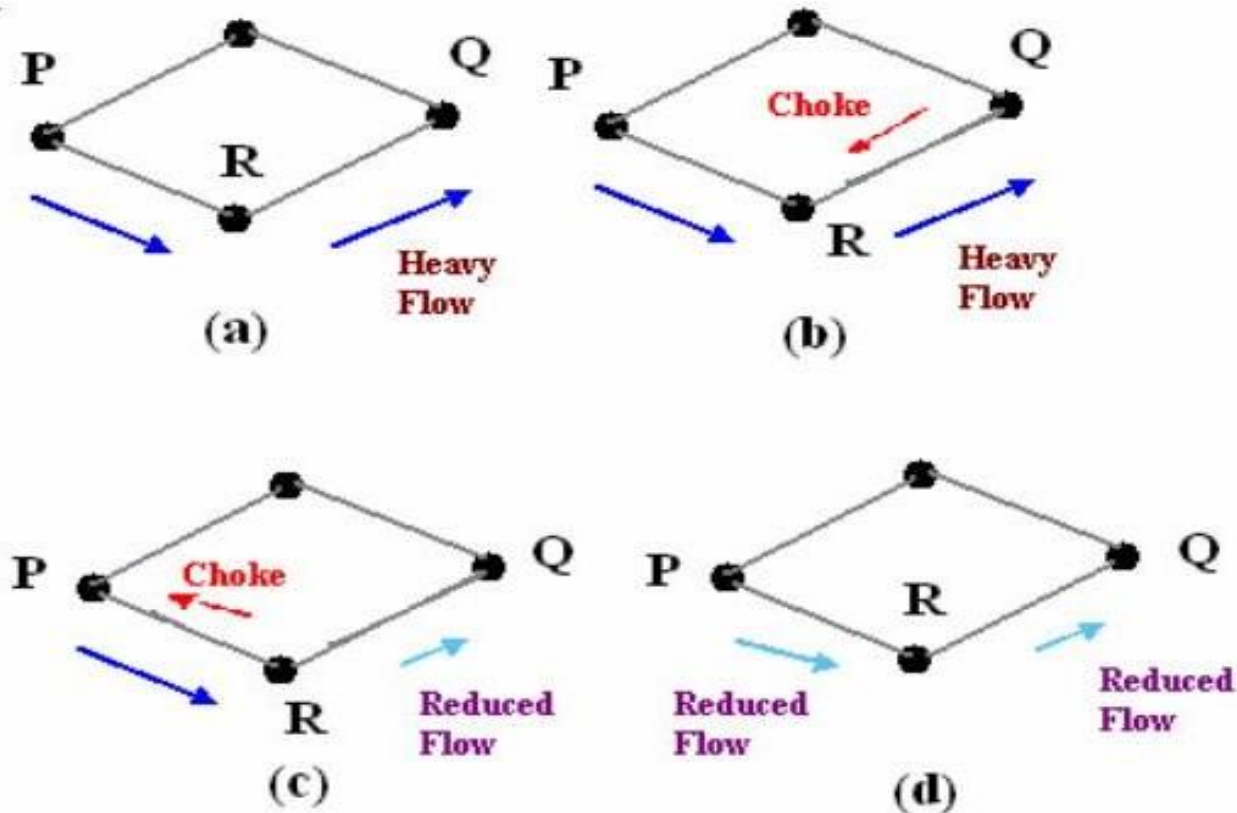


Figure 7.5.7 Depicts the functioning of Hop-by-Hop choke packets, (a) Heavy traffic between nodes P and Q, (b) Node Q sends the Choke packet to P, (c) Choke packet reaches R, and the flow between R and Q is curtailed down, Choke packet reaches P, and P reduces the flow out

Load Shedding

- When none of the above methods, make the congestion disappear, routers can bring out the heavy artillery **Load Shedding**
- Load shedding is a fancy way of saying that when routers are being overloaded by the packets that they cannot handle, they just throw them away.
- The preferred choice of packets to drop may depend on the type of applications that use the network.
- Two strategies normally work: Wine and Milk
- Intelligent load shedding requires cooperation from senders. E.g. control packets (higher priority as compared to data packets), MPEG encoding (i(intra) frames have highest priority than p(predicted) and b(bi directional) frames.
- To implement an intelligent discard policy, applications must mark their packets to indicate to the network how important they are.

Random Early Detection

- Dealing with congestion when it first starts, is more effective than letting it gum up the network and then trying to deal with it.
- By having routers drop packets early, before the situation has become hopeless, there is time for the source to take action before it is too late.
- To determine when to start discarding, routers maintain a running average of their queue lengths.
- When the average queue length or some link exceeds a threshold, the link is said to be congested and a small fraction of packets are discarded at random.
- Picking packets at random makes it more likely that the fastest sender will see a packet drop: this is the best option, since the router cannot tell which source is causing the most trouble in the datagram network.

Quality of Service

- Application requirements
- Traffic shaping
- Packet scheduling
- Admission control
- Integrated services
- Differentiated services

Application Requirements (1)

Application	Bandwidth	Delay	Jitter	Loss
Email	Low	Low	Low	Medium
File sharing	High	Low	Low	Medium
Web access	Medium	Medium	Low	Medium
Remote login	Low	Medium	Medium	Medium
Audio on demand	Low	Low	High	Low
Video on demand	High	Low	High	Low
Telephony	Low	High	High	Low
Videoconferencing	High	High	High	Low

How stringent the quality-of-service requirements are.

Categories of QoS and Examples

a) Constant bit rate

- Telephony

b) Variable bit rate

1. Real-time variable bit rate

- Compressed videoconferencing

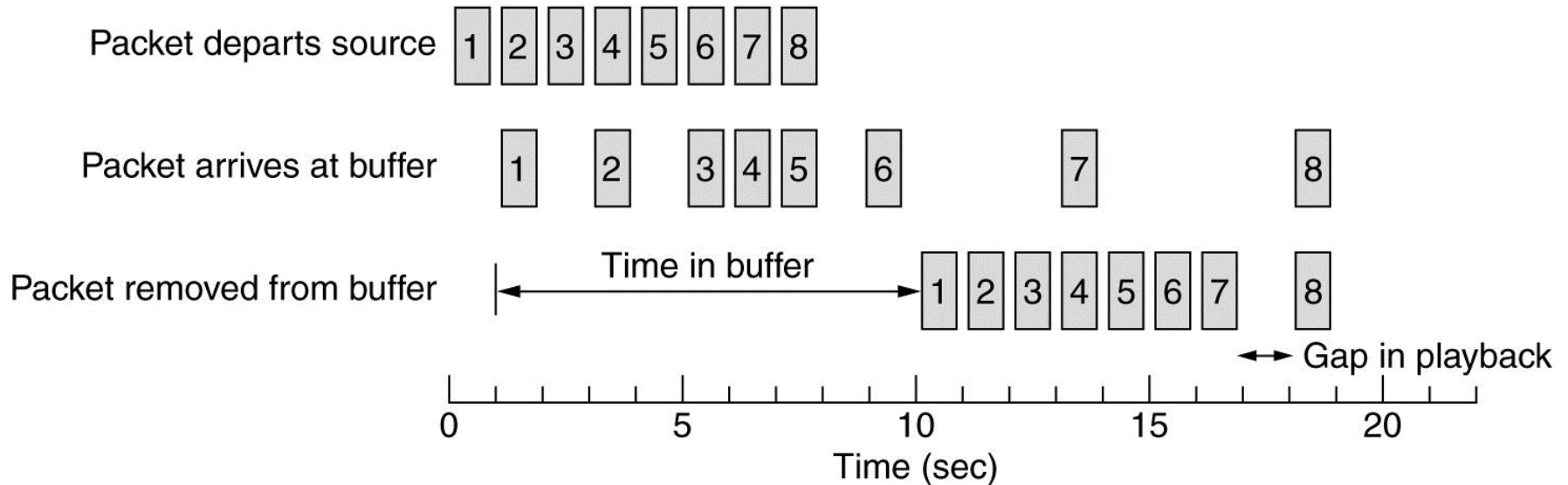
2. Non-real-time variable bit rate

- Watching a movie on demand

c) Available bit rate

- File transfer

Buffering

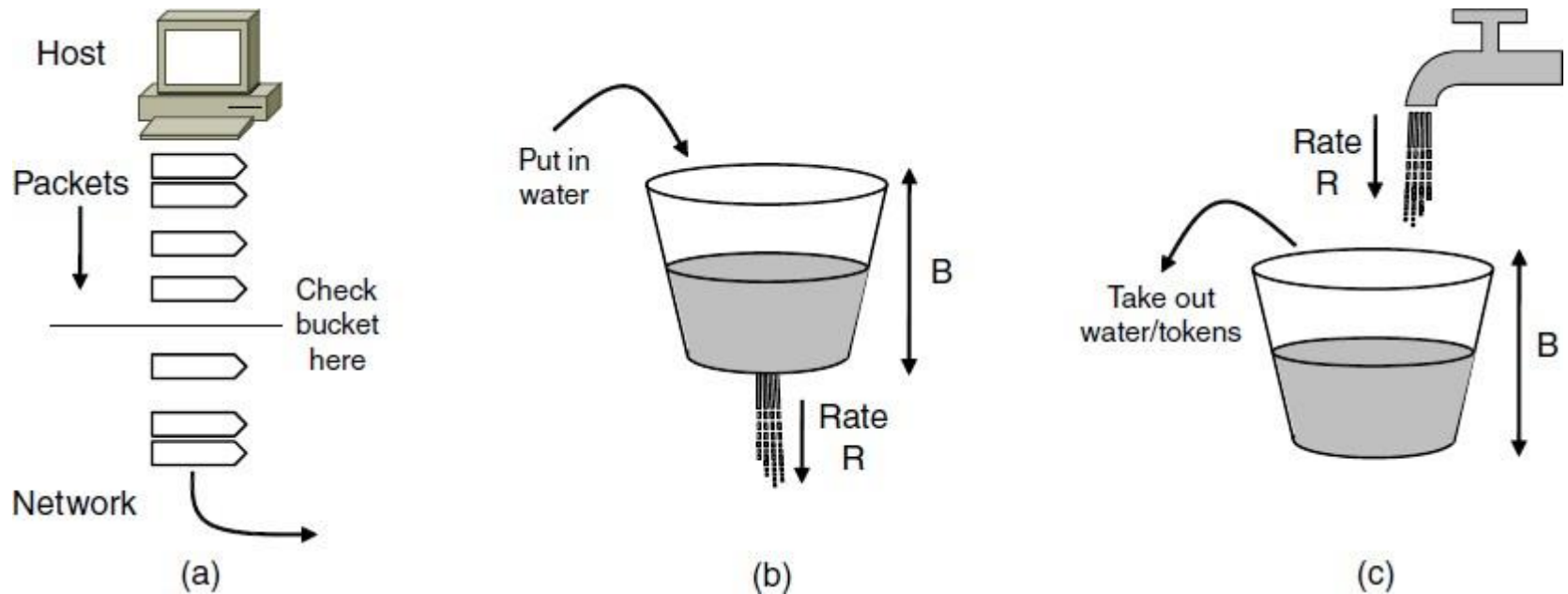


Smoothing the output stream by buffering packets.

Traffic Shaping

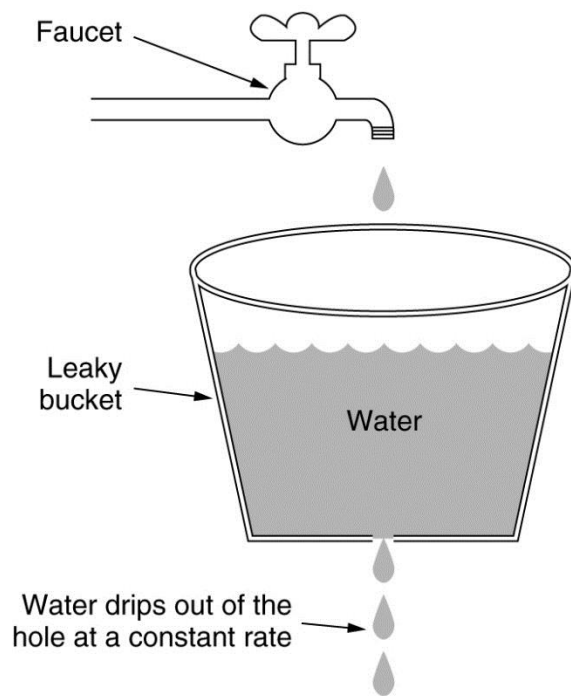
- Before the network can make QoS guarantees, it must know what traffic is being guaranteed
- In CBR, like telephone it is quite easy eg. needs 64 kbps
- **Traffic shaping** is a technique for regulating the average rate and burstiness of a flow of data that enters the network
- It allow applications to transmit a wide variety of traffic that suits their needs, including some bursts, yet have a simple and useful way to describe the possible traffic patterns to the network
- Traffic shaping reduces congestion and thus helps the network live up to its promise
- **SLA (Service Level Agreement)**, is made over aggregate flows and long periods of time
- Monitoring a traffic flow is called **traffic policing**

Traffic Shaping (1)

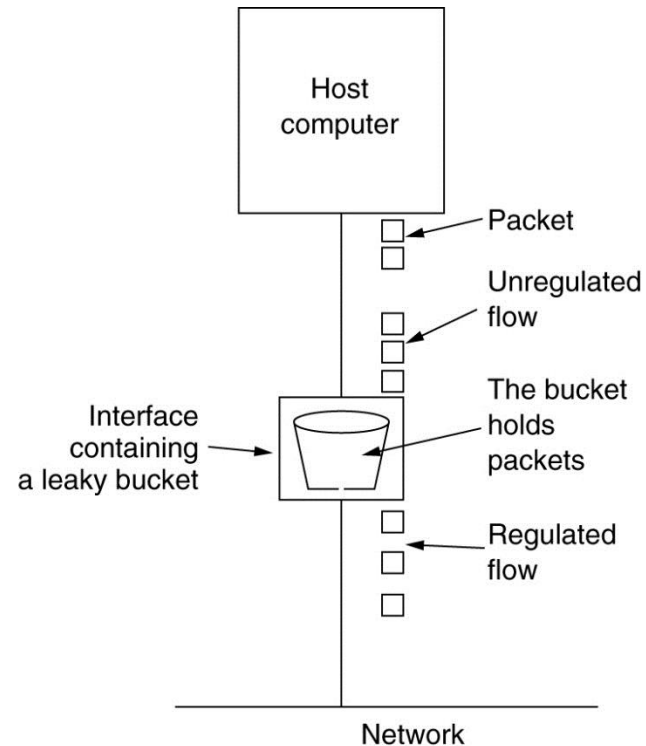


(a) Shaping packets. (b) A leaky bucket. (c) A token bucket

The Leaky Bucket Algorithm



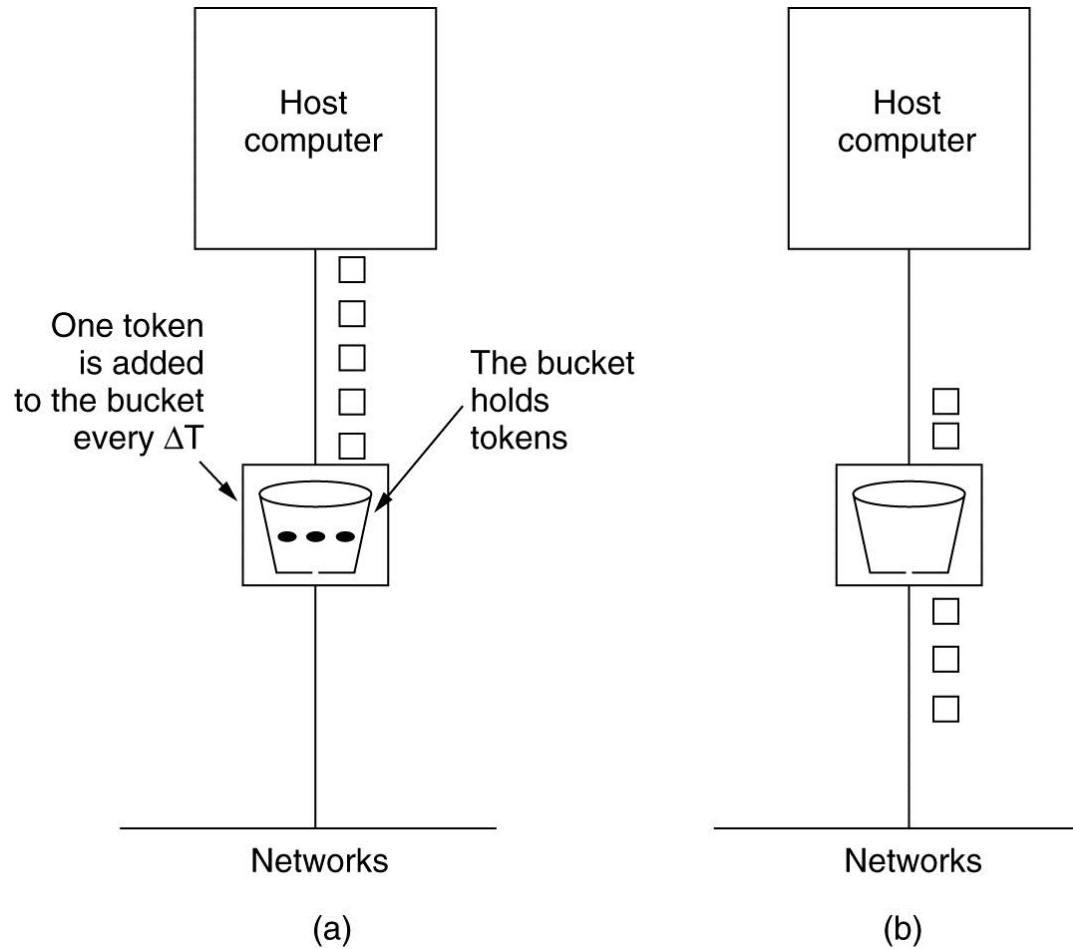
(a)



(b)

(a) A leaky bucket with water. (b) a leaky bucket with packets.

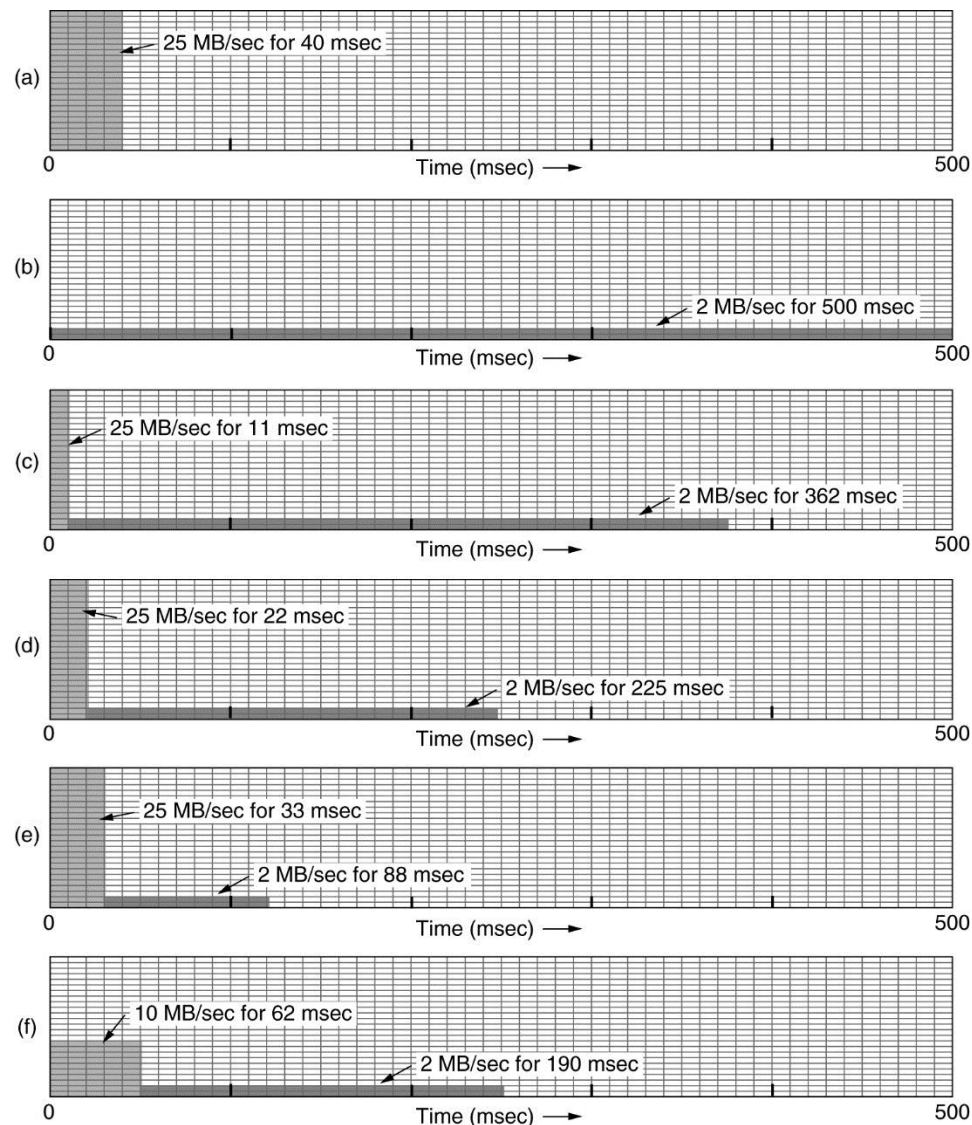
The Token Bucket Algorithm



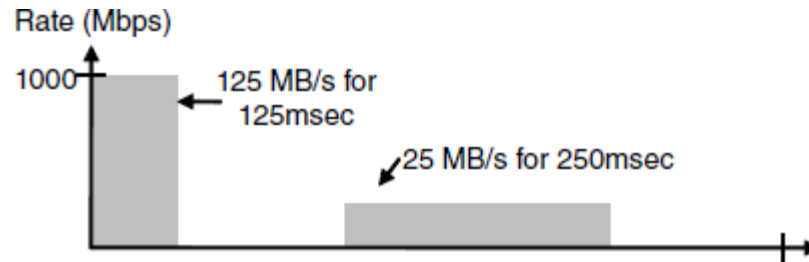
(a) Before. (b) After.

The Leaky Bucket Algorithm

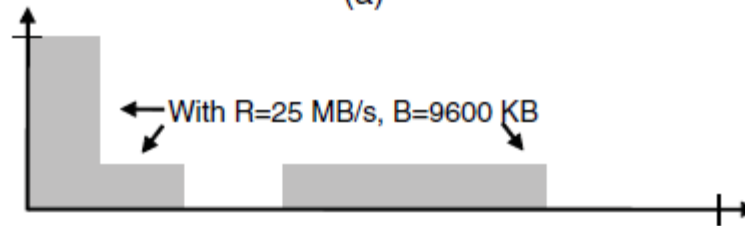
(a) Input to a leaky bucket.
(b) Output from a leaky bucket.
Output from a token bucket with capacities of (c) 250 KB, (d) 500 KB, (e) 750 KB, (f) Output from a 500KB token bucket feeding a 10-MB/sec leaky bucket.



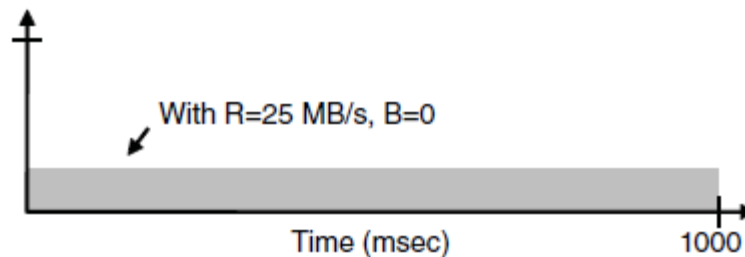
Traffic Shaping (2)



(a)



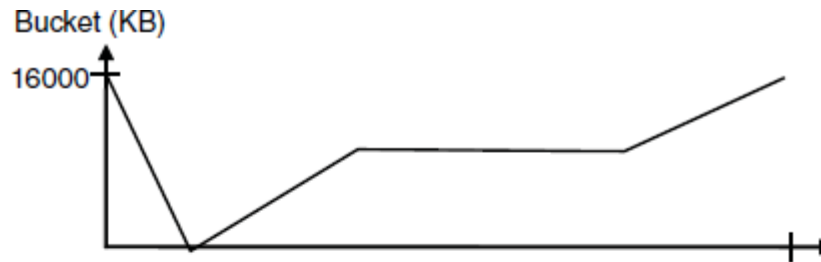
(b)



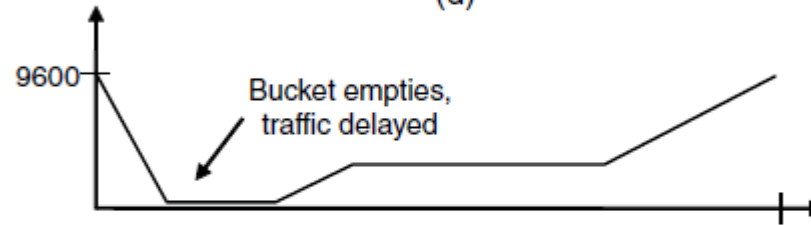
(c)

(a) Traffic from a host. Output shaped by a token bucket of rate 200 Mbps and capacity (b) 9600 KB, (c) 0 KB.

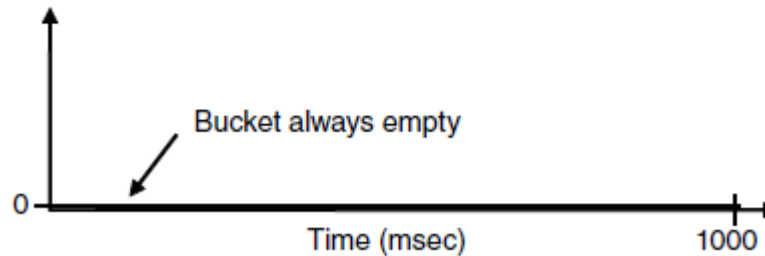
Traffic Shaping (3)



(d)



(e)



(f)

Token bucket level for shaping with rate 200 Mbps and capacity
(d) 16000 KB, (e) 9600 KB, and (f) 0KB..

Traffic Shaping

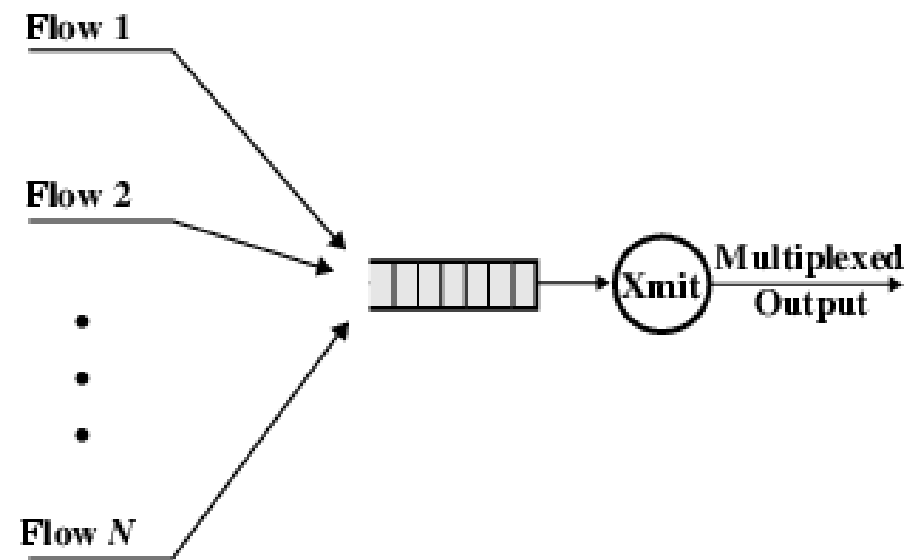
- Calculate the length of the maximum burst
- If burst length is S sec., the maximum output rate M bytes/sec, the token bucket capacity B bytes, and the token arrival rate R bytes/sec, the output burst contains a maximum of $B + RS$ bytes which gets transferred at M bytes/sec for S sec
- $B + RS = MS$ or $S = B/(M-R)$

Packet Scheduling

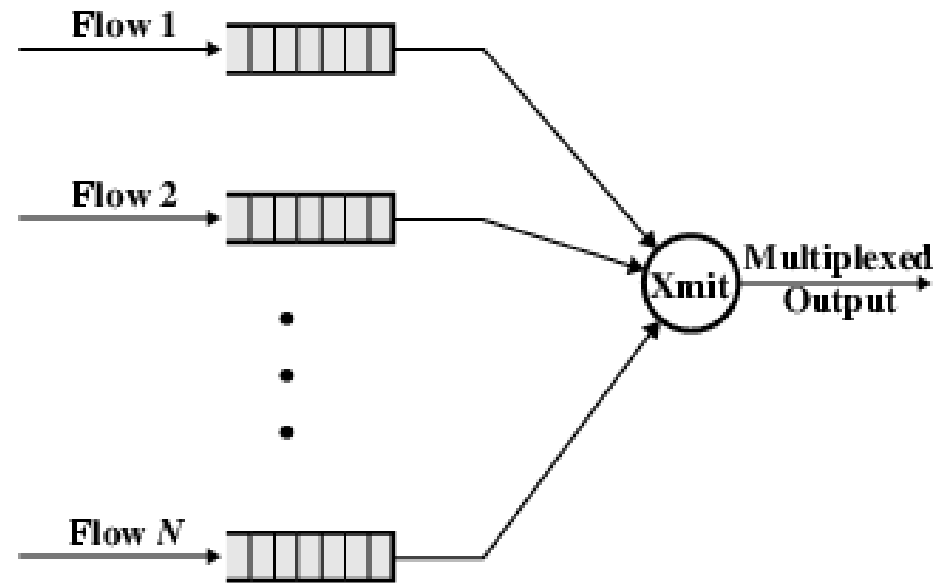
Kinds of resources can potentially be reserved for different flows:

- Bandwidth.
- Buffer space.
- CPU cycles.

Packet Scheduling

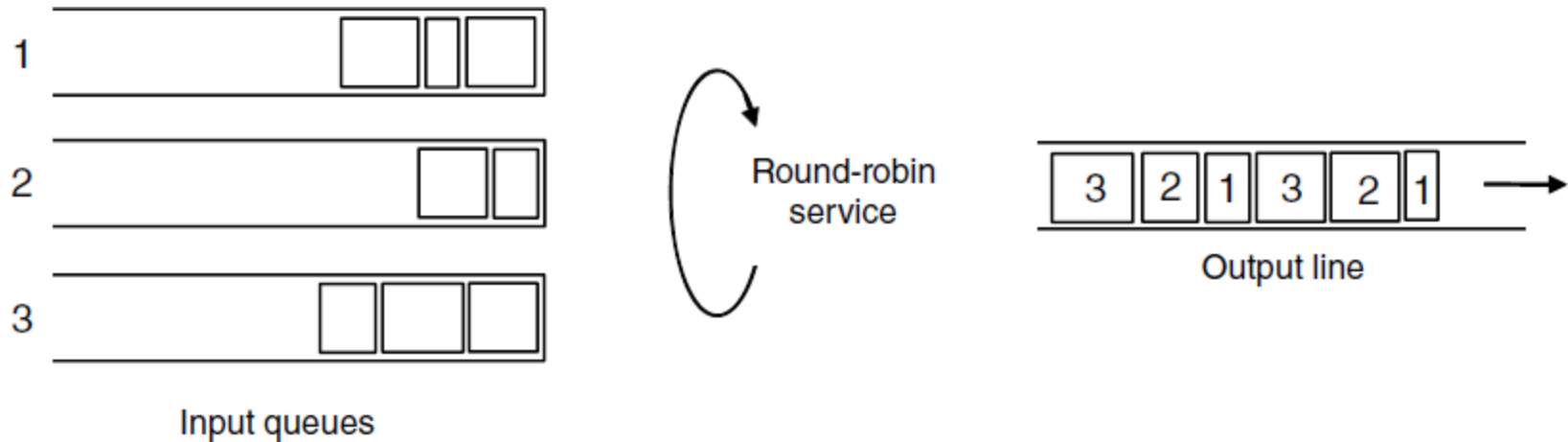


(a) FIFO Queuing



(b) Fair Queuing

Packet Scheduling



Round-robin Fair Queuing

Internetworking

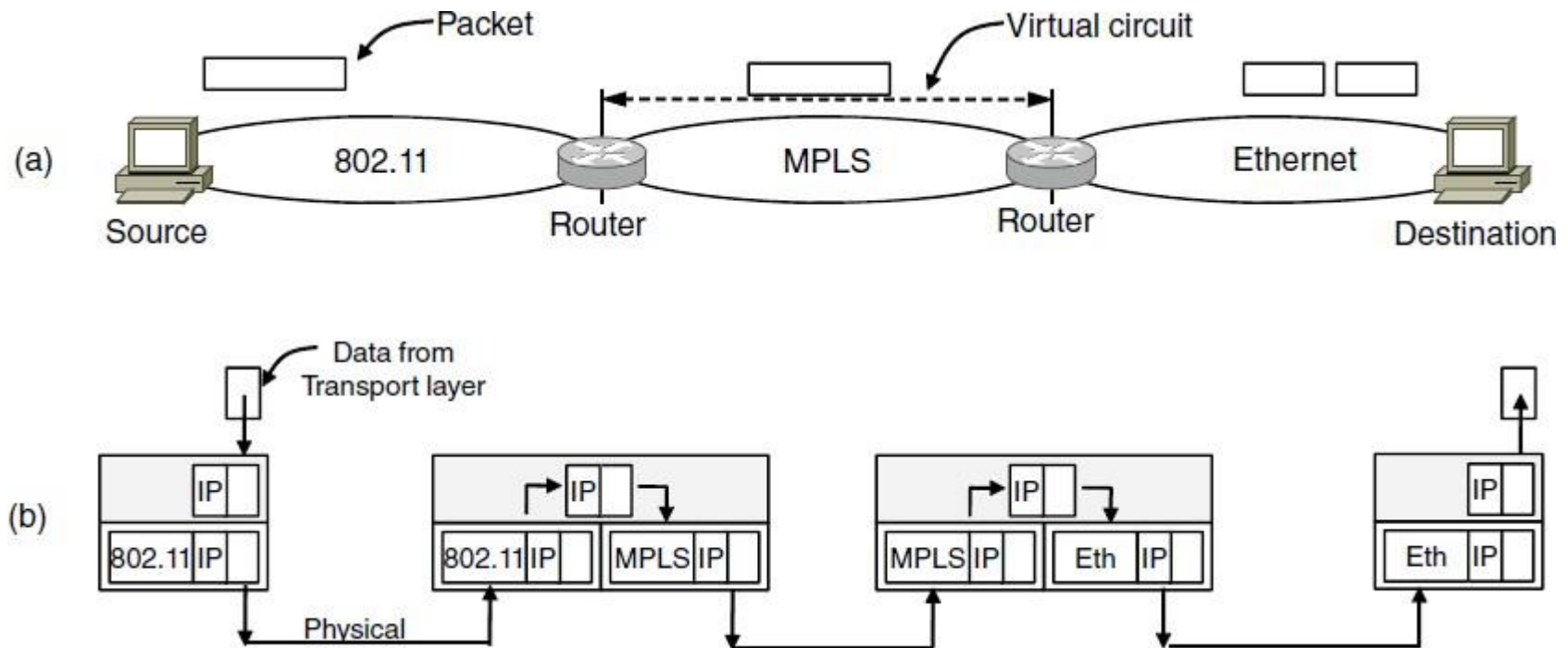
- How networks differ
- How networks can be connected
- Tunneling
- Internetwork routing
- Packet fragmentation

How Networks Differ

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

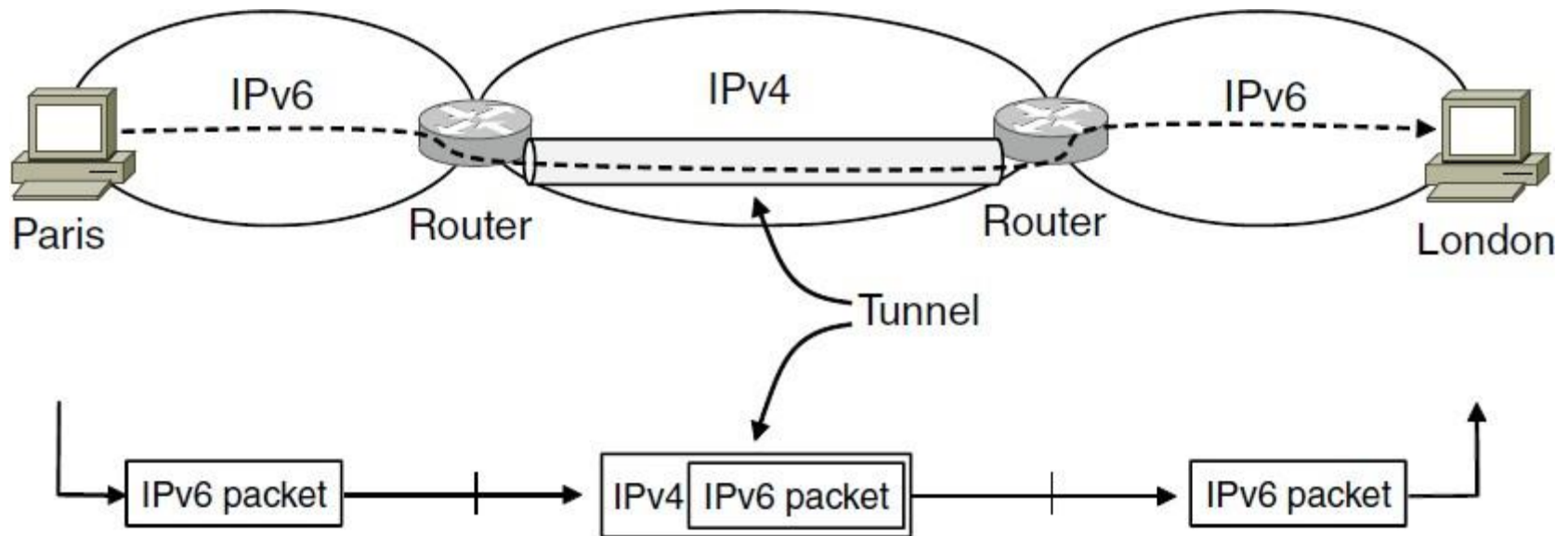
Some of the many ways networks can differ

How Networks Can Be Connected



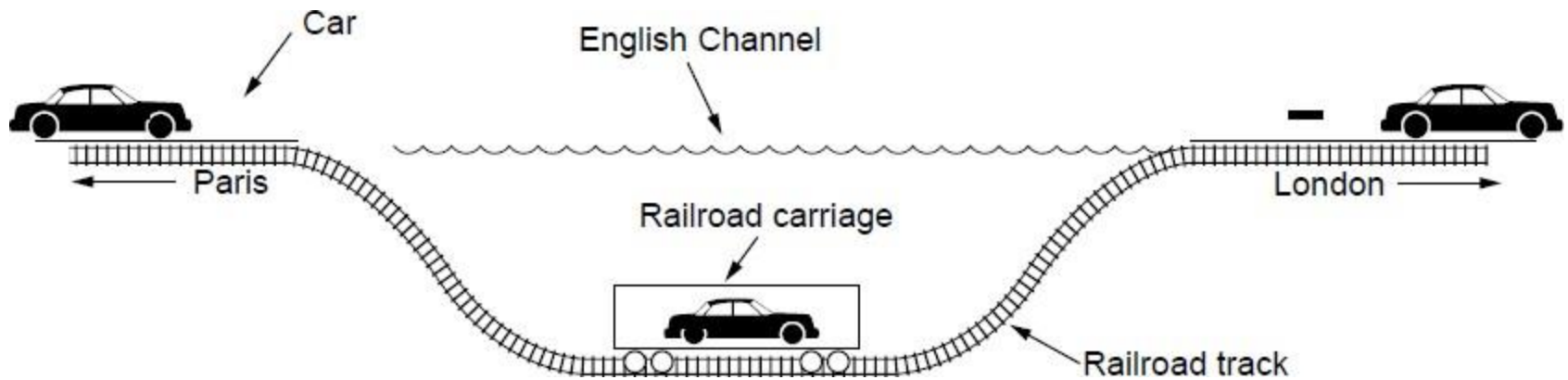
- a) A packet crossing different networks.
- b) Network and link layer protocol processing.

Tunneling (1)



Tunneling a packet from Paris to London.

Tunneling (2)



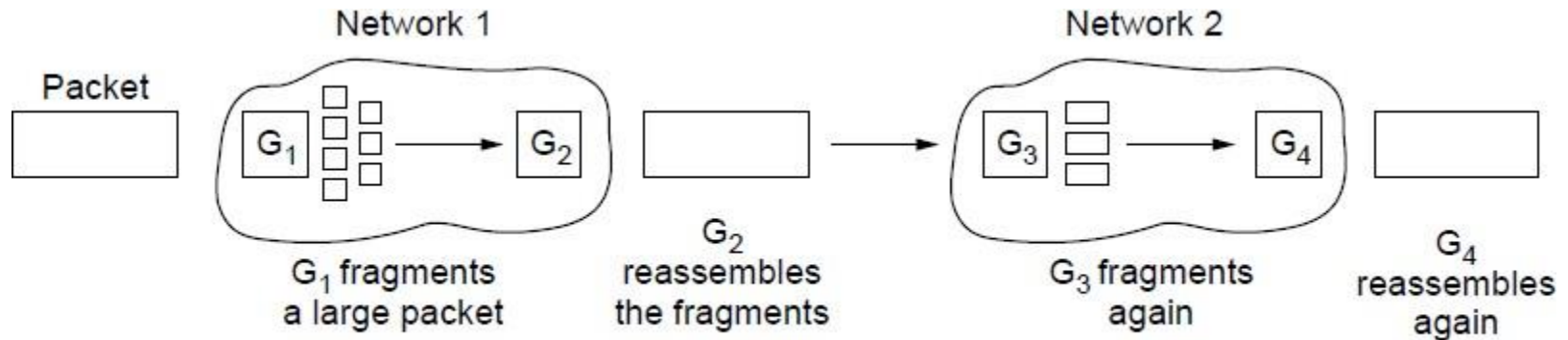
Tunneling a car from France to England

Packet Fragmentation (1)

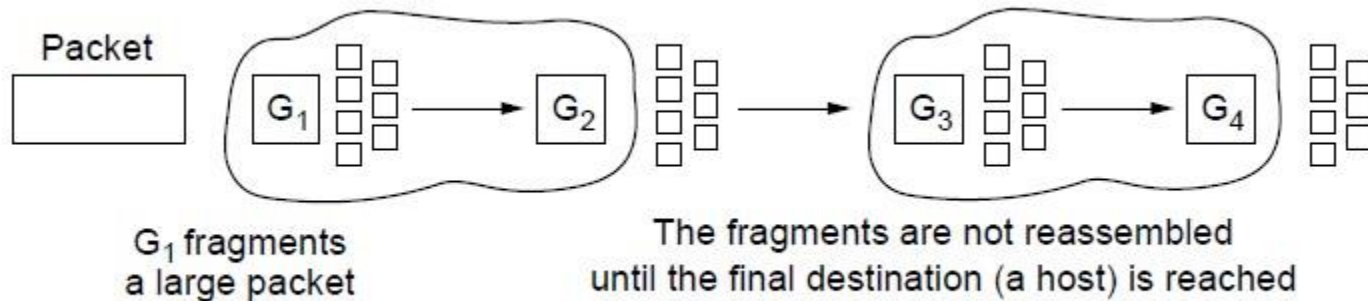
Packet size issues:

- Hardware
- Operating system
- Protocols
- Compliance with (inter)national standard.
- Reduce error-induced retransmissions
- Prevent packet occupying channel too long.

Packet Fragmentation (2)



(a)

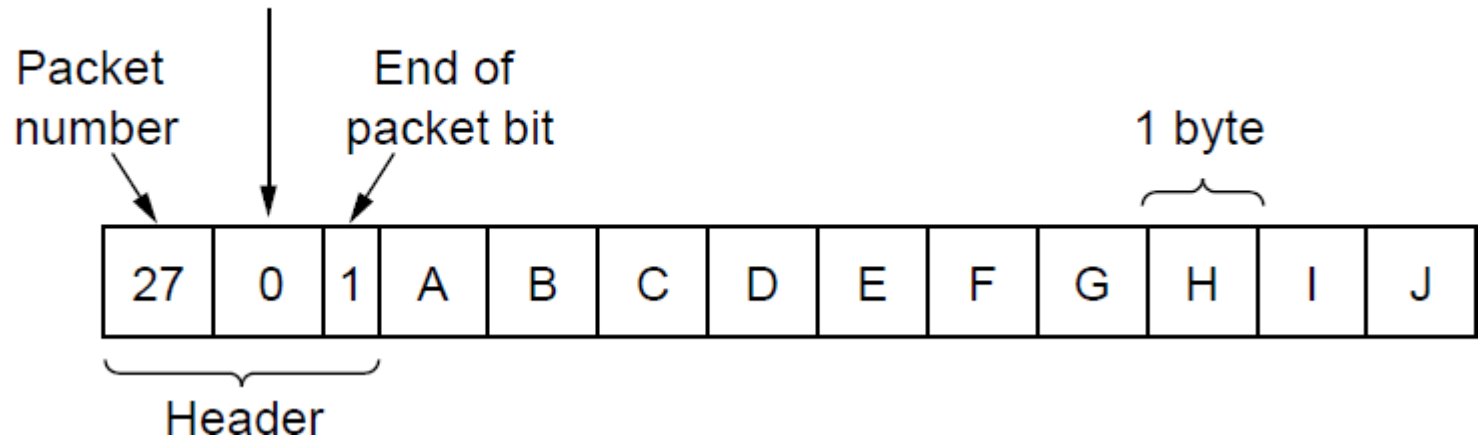


(b)

- a) Transparent fragmentation.
- b) Nontransparent fragmentation

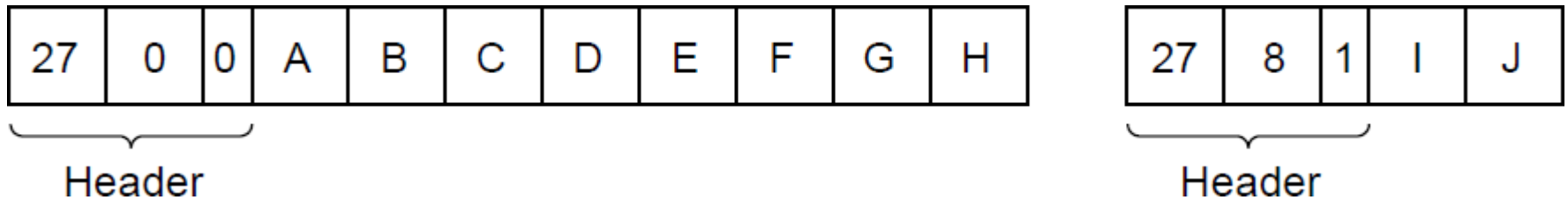
Packet Fragmentation (3)

Number of the first elementary fragment in this packet



Fragmentation when the elementary data size is 1 byte.
(a) Original packet, containing 10 data bytes.

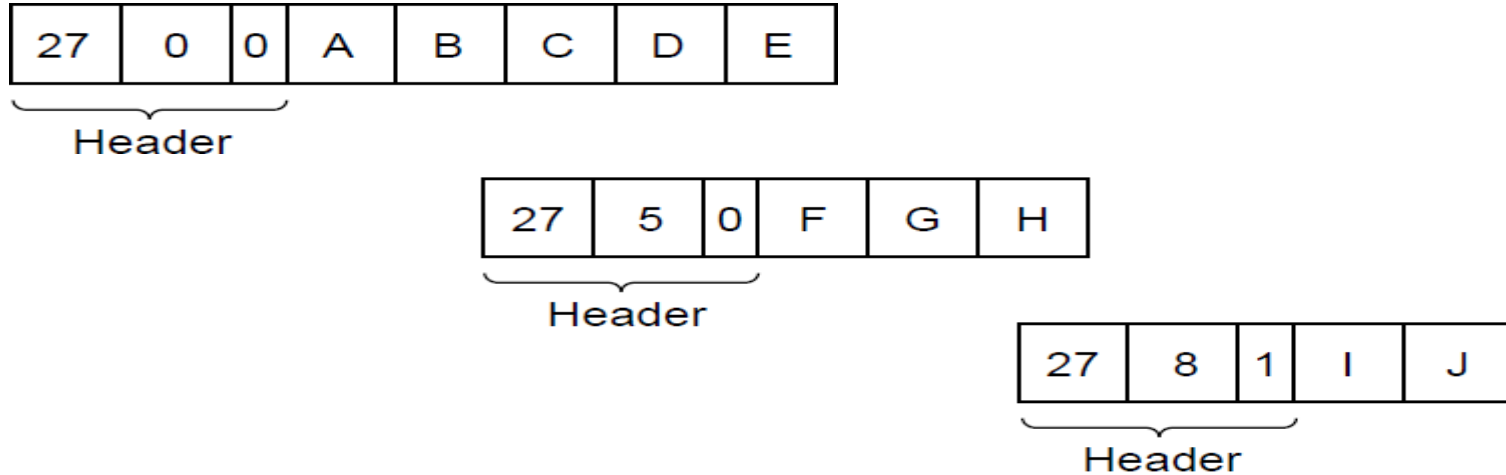
Packet Fragmentation (4)



Fragmentation when the elementary data size is 1 byte

(b) Fragments after passing through a network
with maximum packet size of 8 payload bytes plus header.

Packet Fragmentation (5)

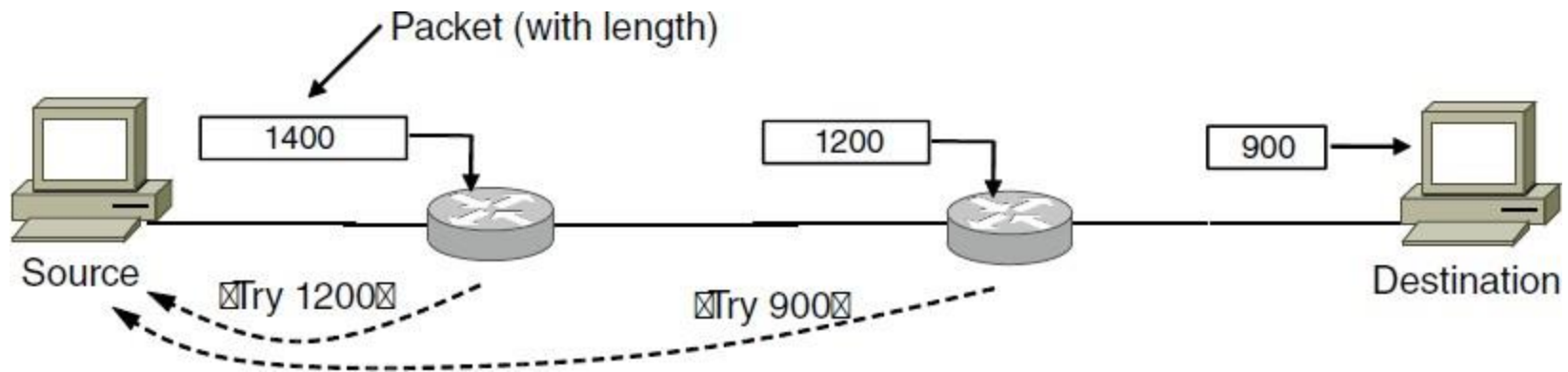


Fragmentation when the elementary data size is 1 byte

(c) Fragments after passing through a size 5 gateway.

In IPv6 all the routers cannot do fragmentation, only the source can do, whereas in IPv4 any router can do fragmentation.

Packet Fragmentation (6)



Path MTU Discovery

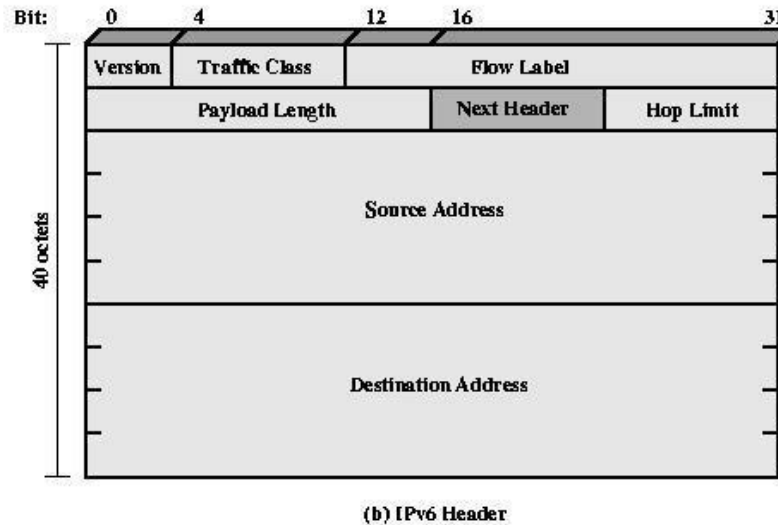
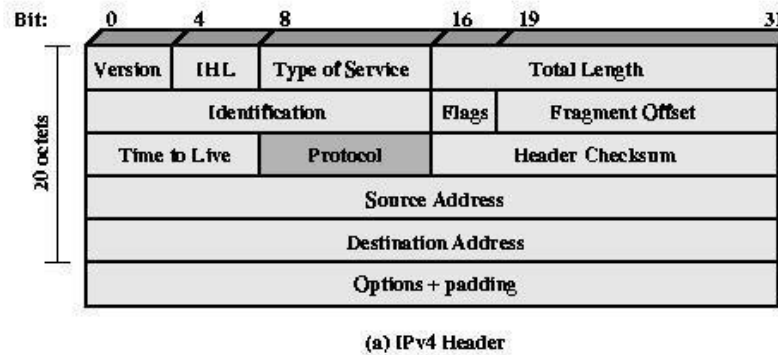


Figure 2.2 IP Headers

Fragmentation and Reassembly

- a) Networks may have different maximum packet size
- b) Router may need to fragment datagrams before sending to next network
- c) Fragments may need further fragmenting in later networks
- d) Reassembly done only at final destination since fragments may take different routes

Fragmentation and Reassembly

- a) If Reassembly permitted at intermediate routers
- Large buffers needed to store partial datagram
 - All fragment must pass through same gateway

Fragmentation Example

MTUs for some networks

<i>Protocol</i>	<i>MTU</i>
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

Fragmentation Example

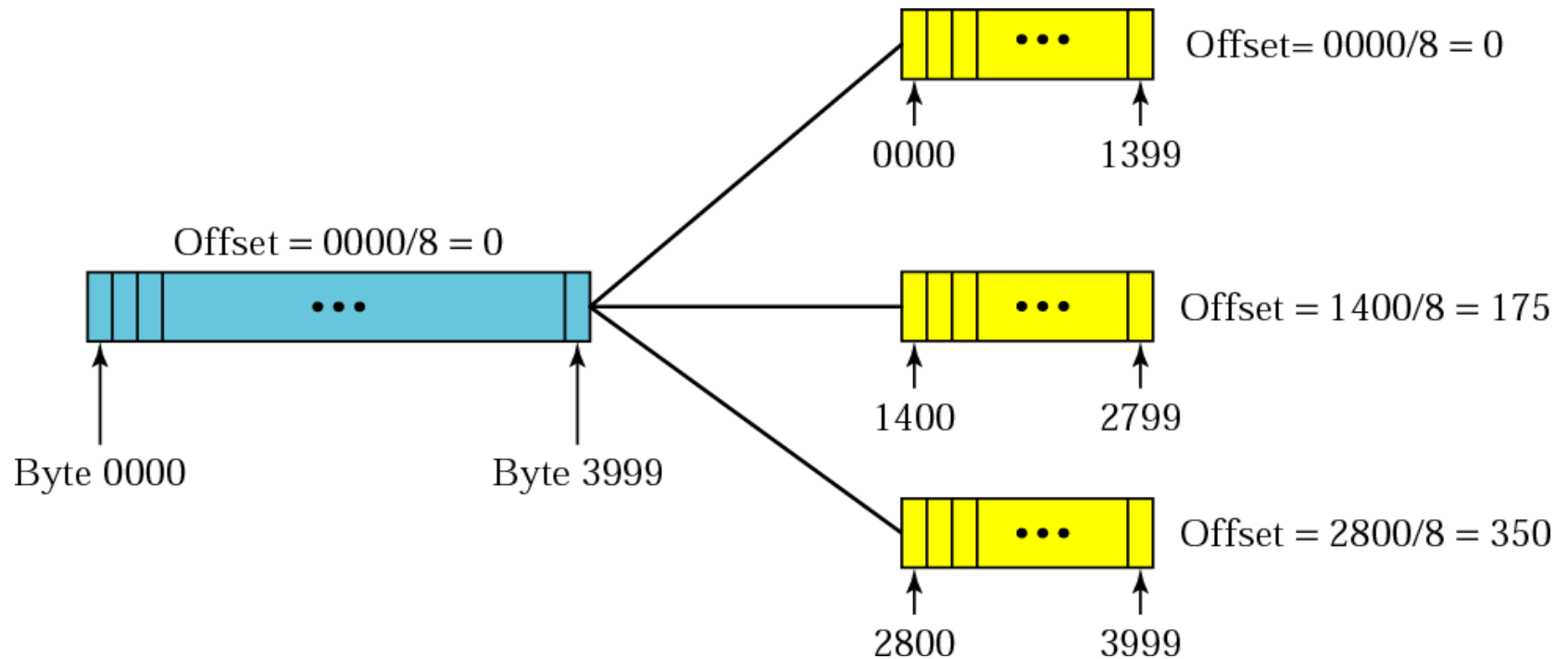
Flags field

D: Do not fragment

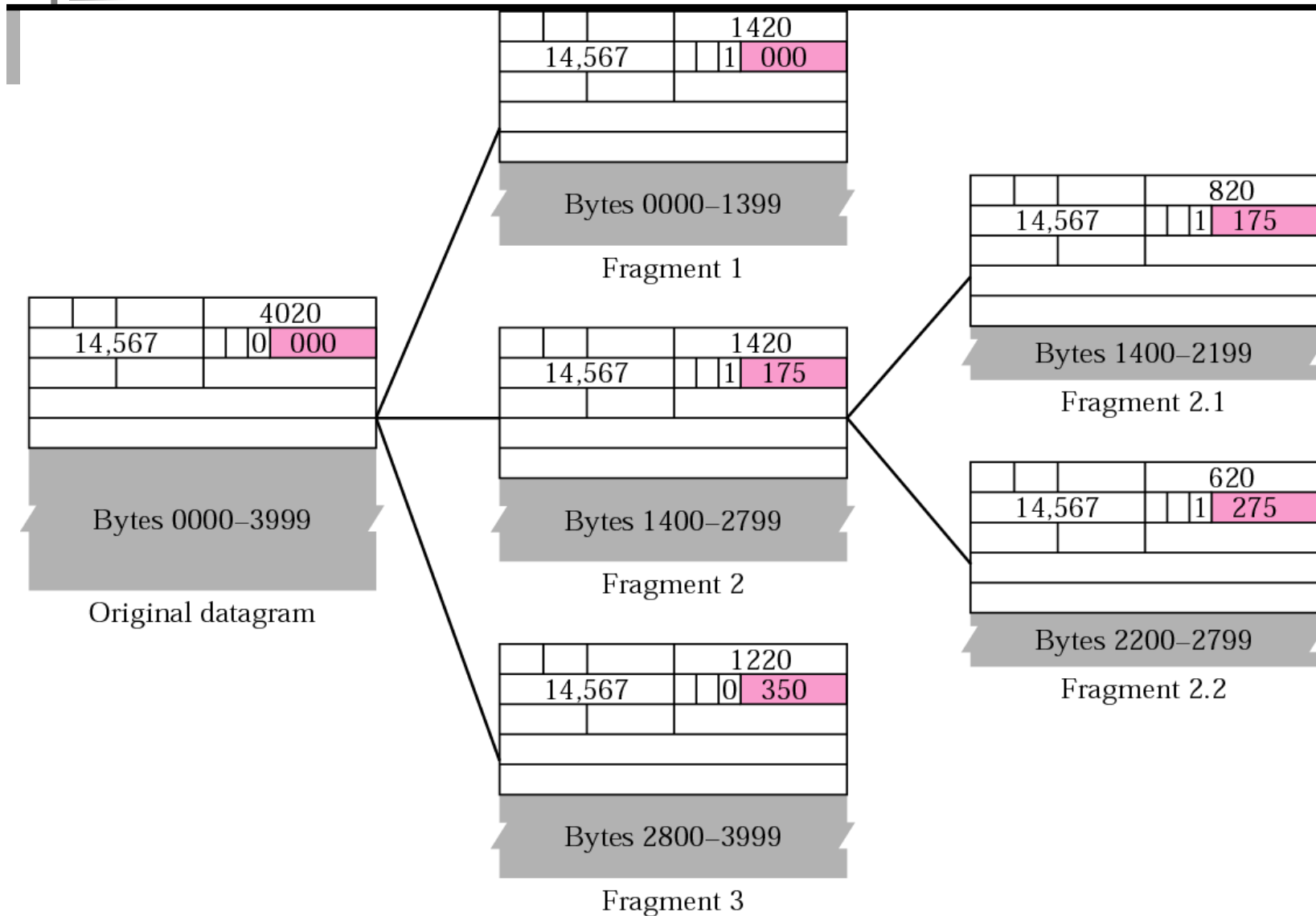
M: More fragments



Fragmentation Example



Fragmentation Example



The Network Layer Principles (1)

- Make sure it works
- Keep it simple
- Make clear choices
- Exploit modularity
- Expect heterogeneity
- . . .

The Network Layer Principles (2)

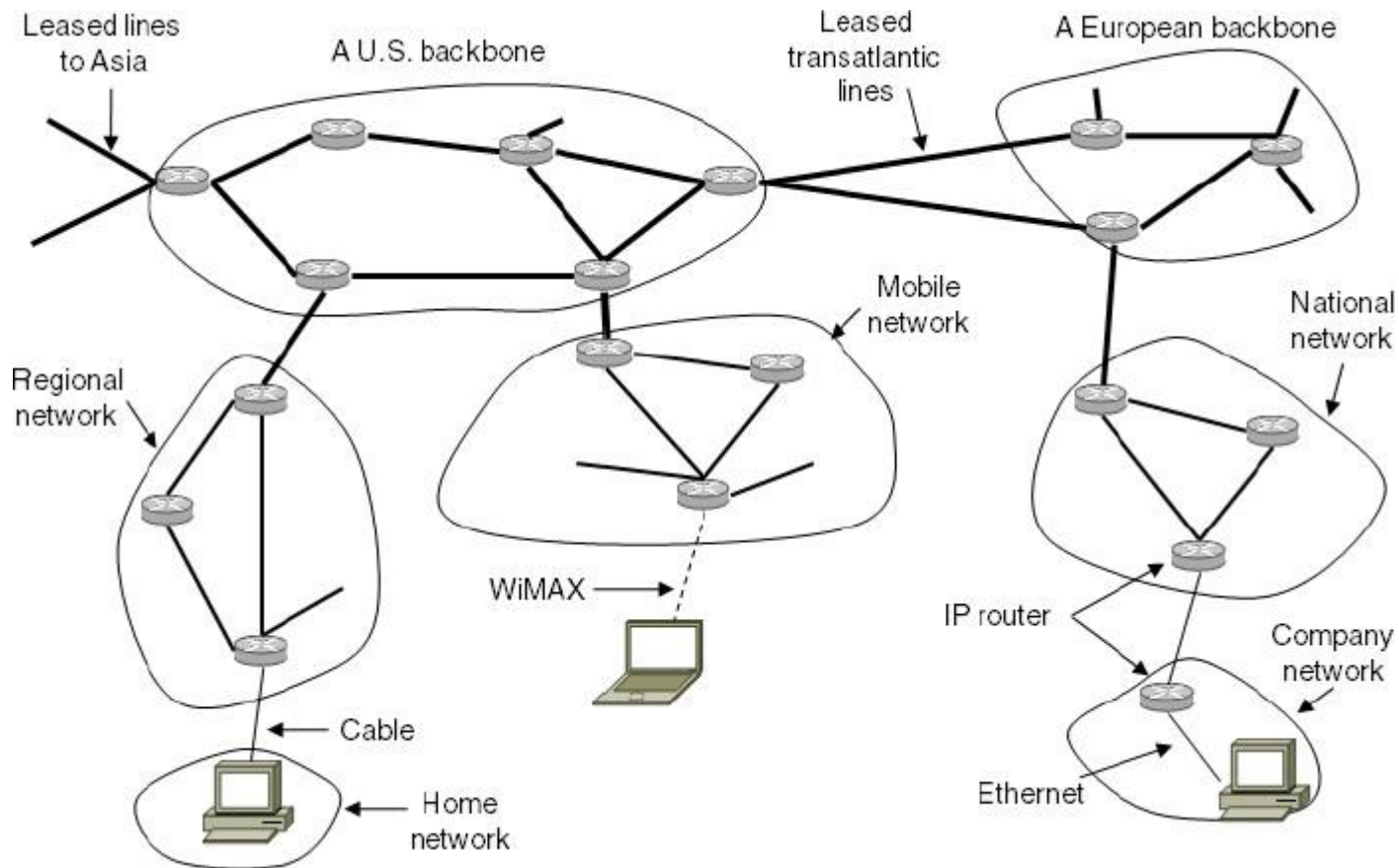
...

- Avoid static options and parameters
- Look for good design (not perfect)
- Strict sending, tolerant receiving
- Think about scalability
- Consider performance and cost

The Network Layer in the Internet (1)

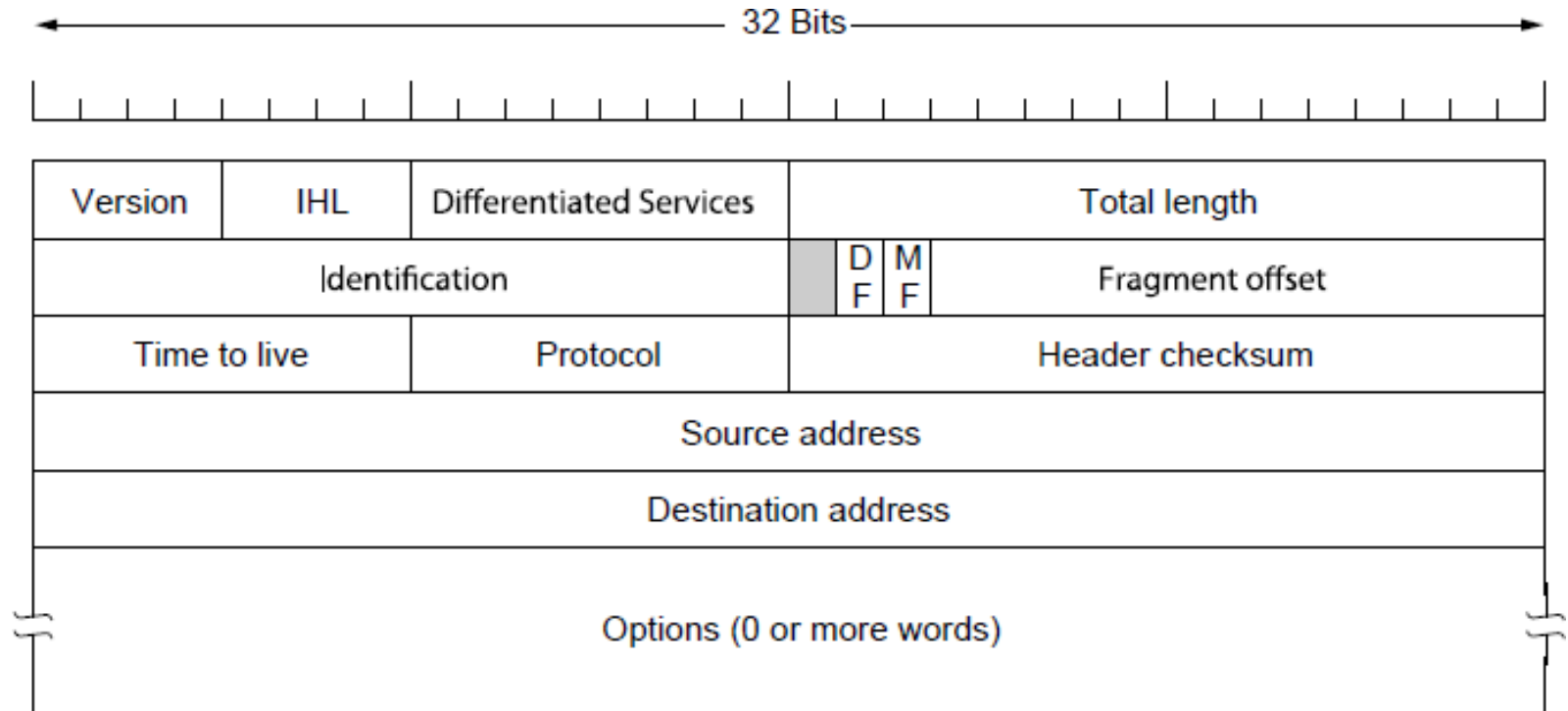
- The IP Version 4 Protocol
- IP Addresses
- IP Version 6
- Internet Control Protocols
- Label Switching and MPLS
- OSPF—An Interior Gateway Routing Protocol
- BGP—The Exterior Gateway Routing Protocol
- Internet Multicasting
- Mobile IP

The Network Layer in the Internet (2)



The Internet is an interconnected collection of many networks.

The IP Version 4 Protocol (1)



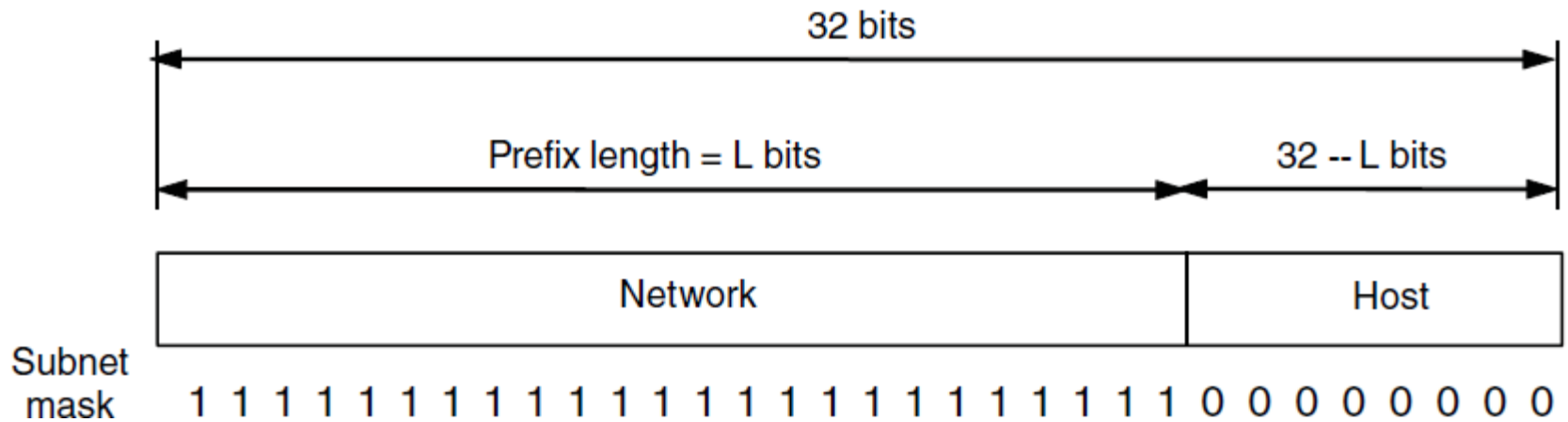
The IPv4 (Internet Protocol) header.

The IP Version 4 Protocol (2)

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

Some of the IP options.

IP Addresses (1)

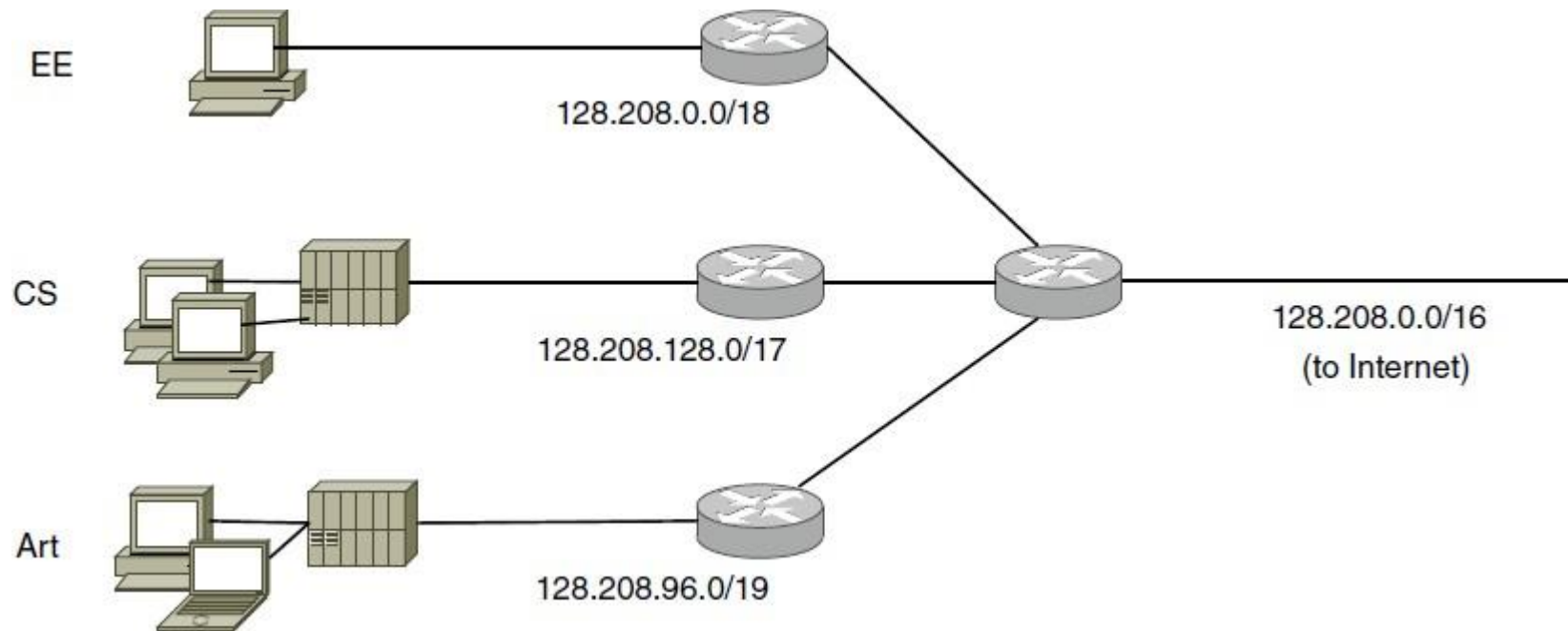


An IP prefix.

Summary of IP Addresses

CLASS	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORK	START ADDRESS	END ADDRESS
CLASS A	0	8	24	2^7 (128)	2^{24} (16,777,216)	0.0.0.0	127.255.255.255
CLASS B	10	16	16	2^{14} (16,384)	2^{16} (65,536)	128.0.0.0	191.255.255.255
CLASS C	110	24	8	2^{21} (2,097,152)	2^8 (256)	192.0.0.0	223.255.255.255
CLASS D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	224.0.0.0	239.255.255.255
CLASS E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	240.0.0.0	255.255.255.255

IP Addresses (2)



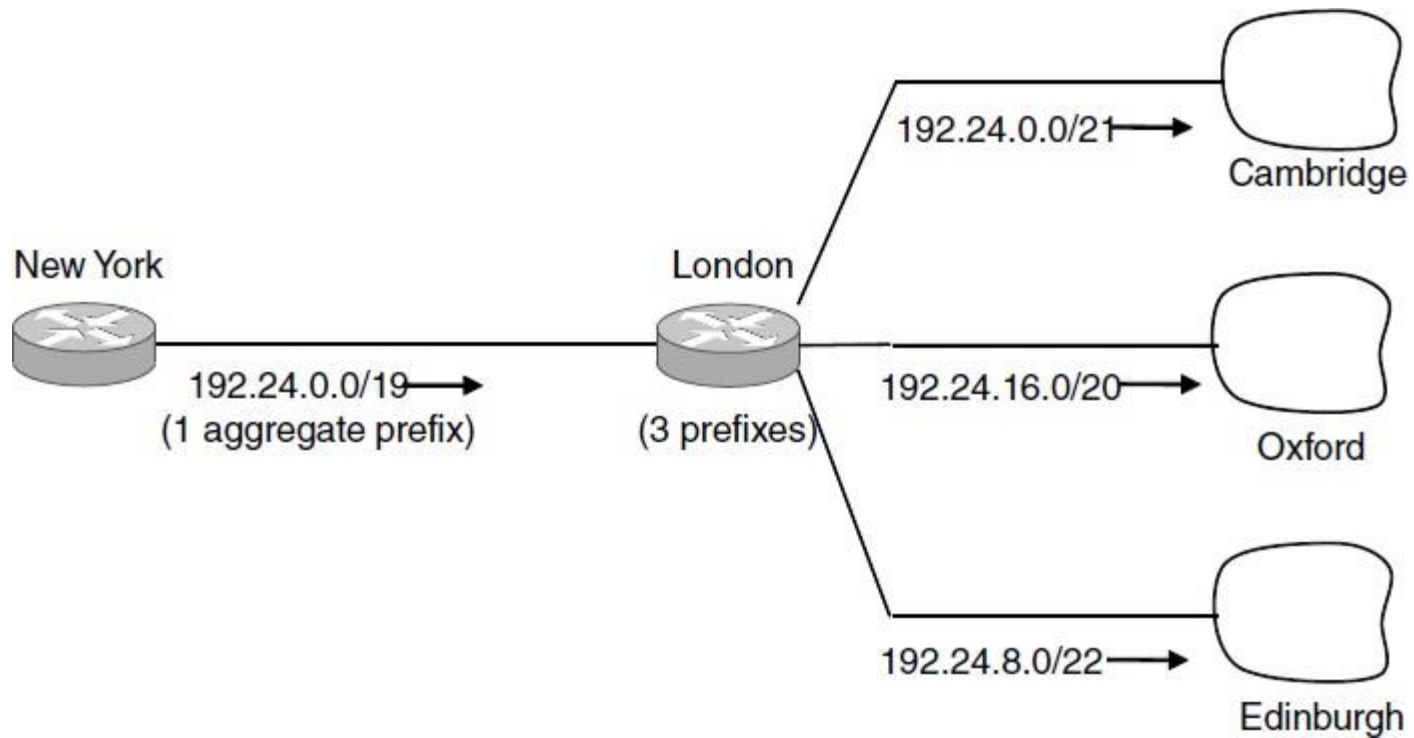
Splitting an IP prefix into separate networks with subnetting.

IP Addresses (3)

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

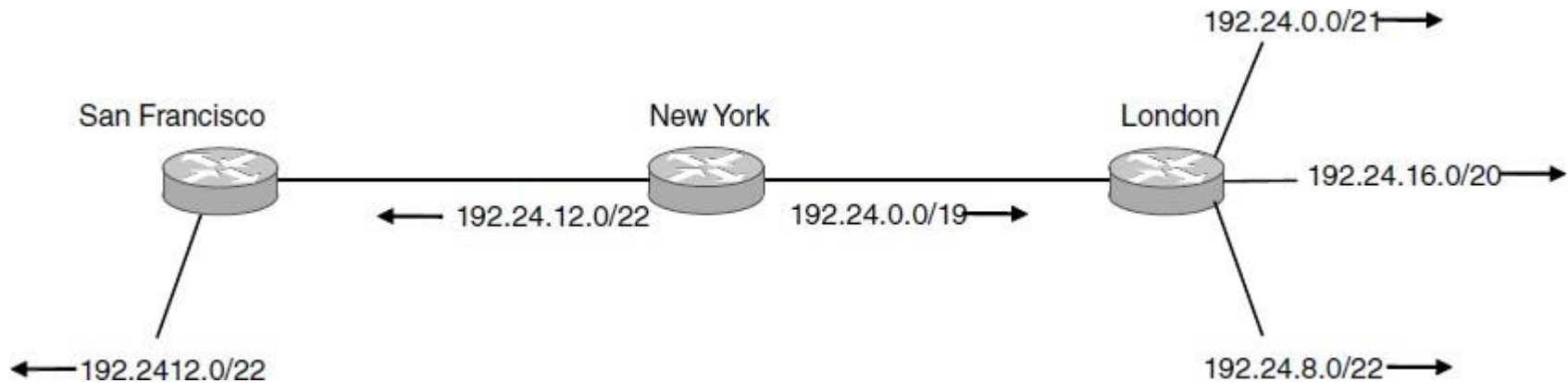
A set of IP address assignments

IP Addresses (4)



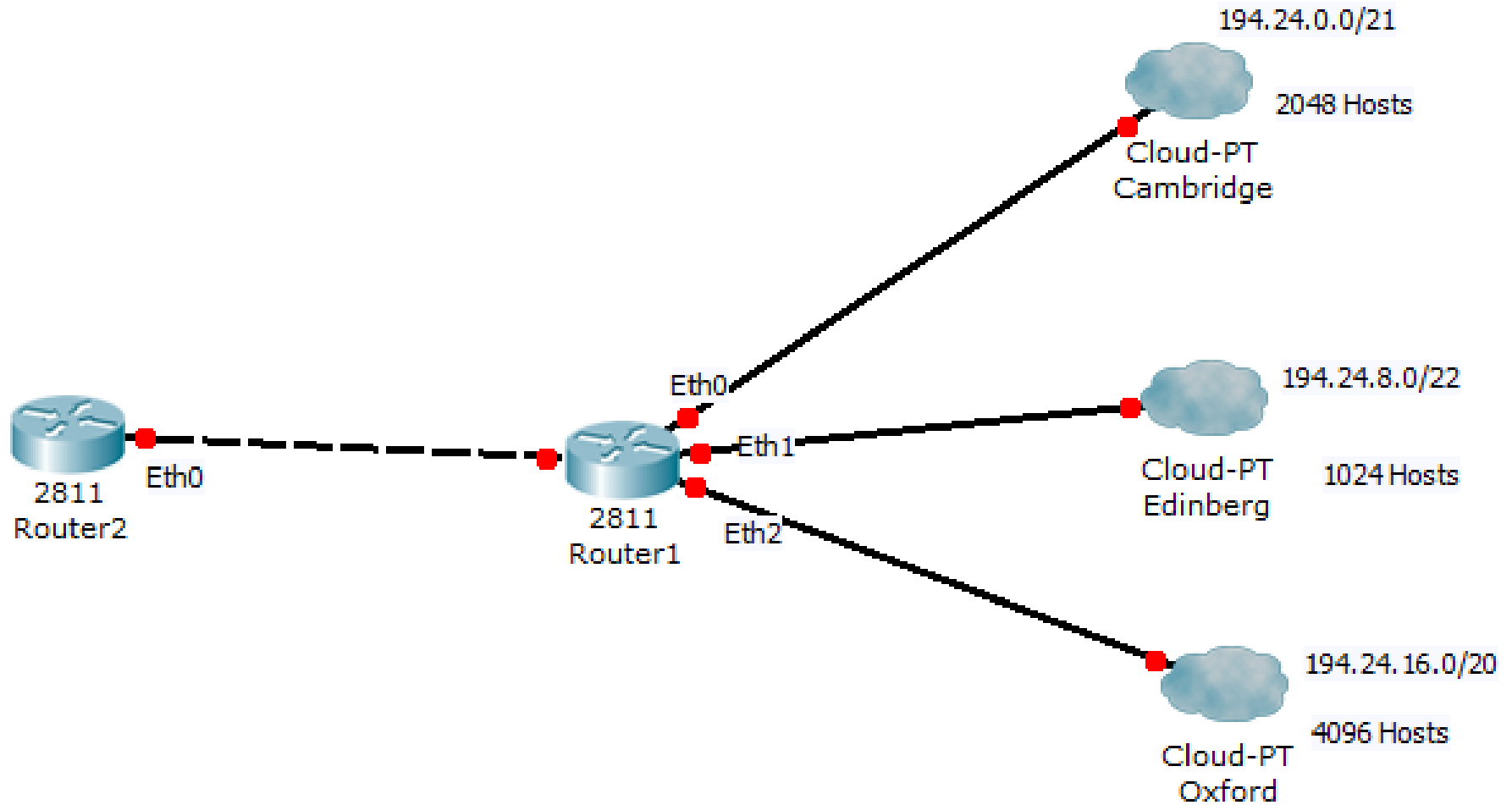
Aggregation of IP prefixes

IP Addresses (5)



Longest matching prefix routing at the New York router.

Network



CDR – Classless InterDomain Routing

Routing Table
at Router 1

	Address	Mask	Interface
Cambridge	194.24.0.0	21	Eth0
Edinberg	194.24.8.0	22	Eth1
Oxford	194.24.16.0	20	Eth2

Consider a packet with 194.24.17.4. Compare to all the entries and use longest matching entry

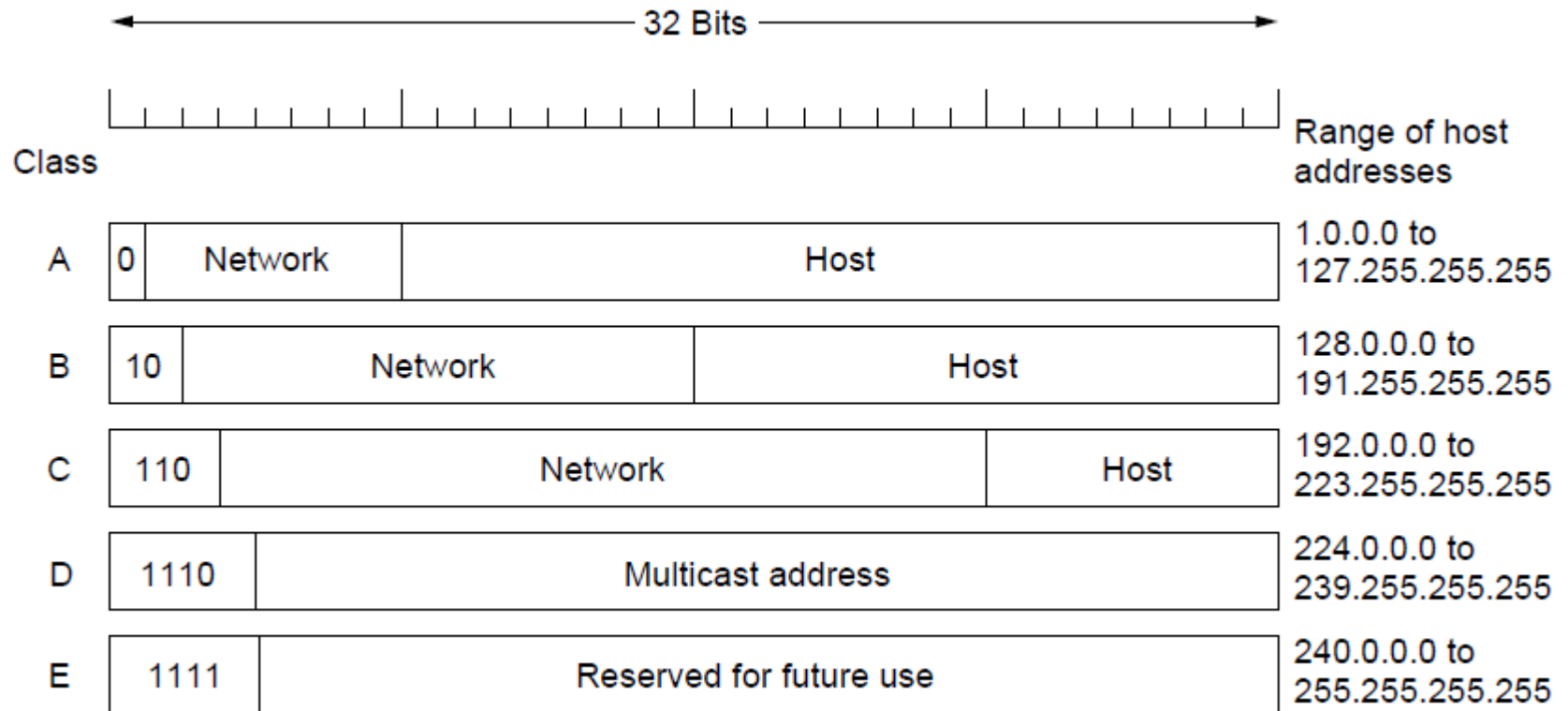
Routing Table
at Router 2

	Address	Mask	Interface
Cambridge	194.24.0.0	21	Eth0
Edinberg	194.24.8.0	22	Eth0
Oxford	194.24.16.0	20	Eth0

Aggregated Routing
Table at Router 2

	Address	Mask	Interface
All	194.24.0.0	19	Eth0

IP Addresses (6)



IP address formats

IP Addresses (7)

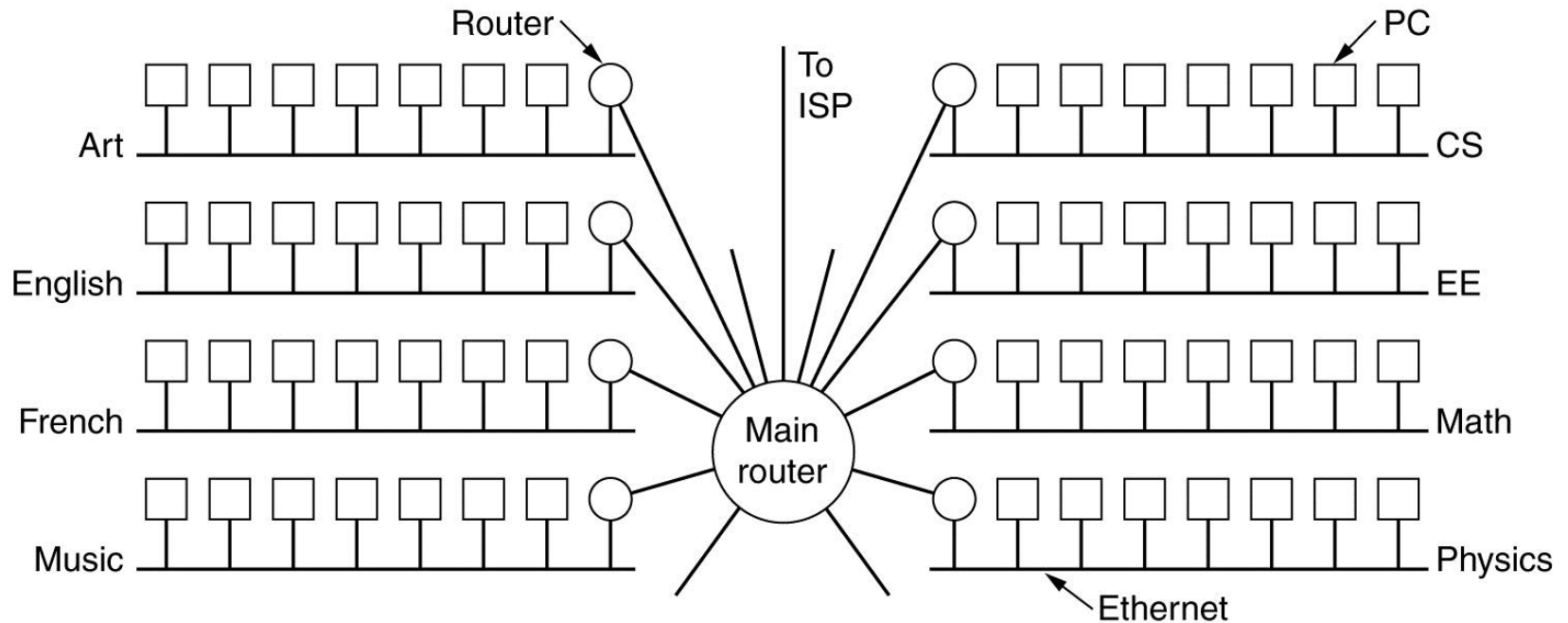
0 0																														This host
0 0				...				0 0				Host																		A host on this network
1 1																														Broadcast on the local network
Network								1 1 1 1				...				1 1 1 1				Broadcast on a distant network										
127				(Anything)																										Loopback

Special IP addresses

IP Addressing

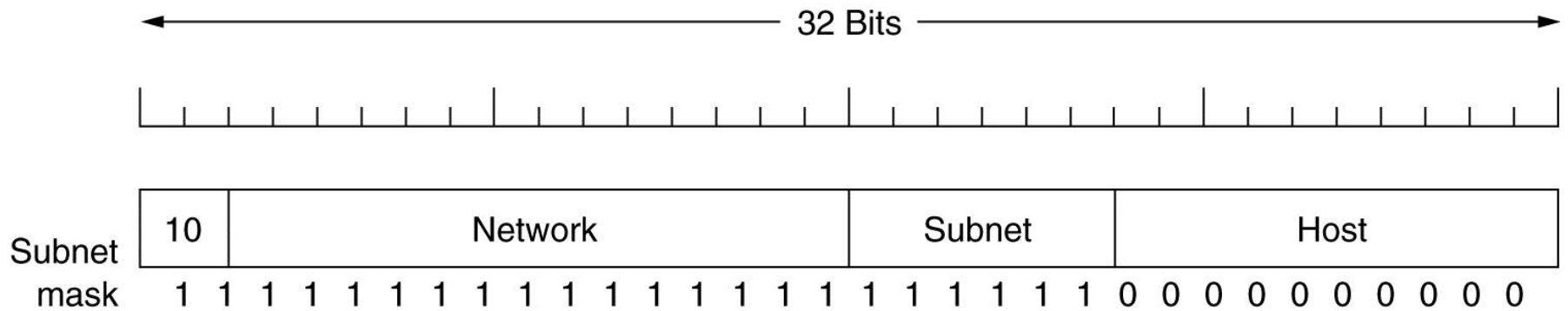
Given the network address 133.22.0.0, find the class, the network id, and the range of the addresses.

Subnets



A campus network consisting of LANs for various departments.

Subnets (2)



A class B network subnetted into 64 subnets.

Subnetting

What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?

Subnetting

1. A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets. Find the number of addresses of each subnet.
2. A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?

The IPv4 Shortage

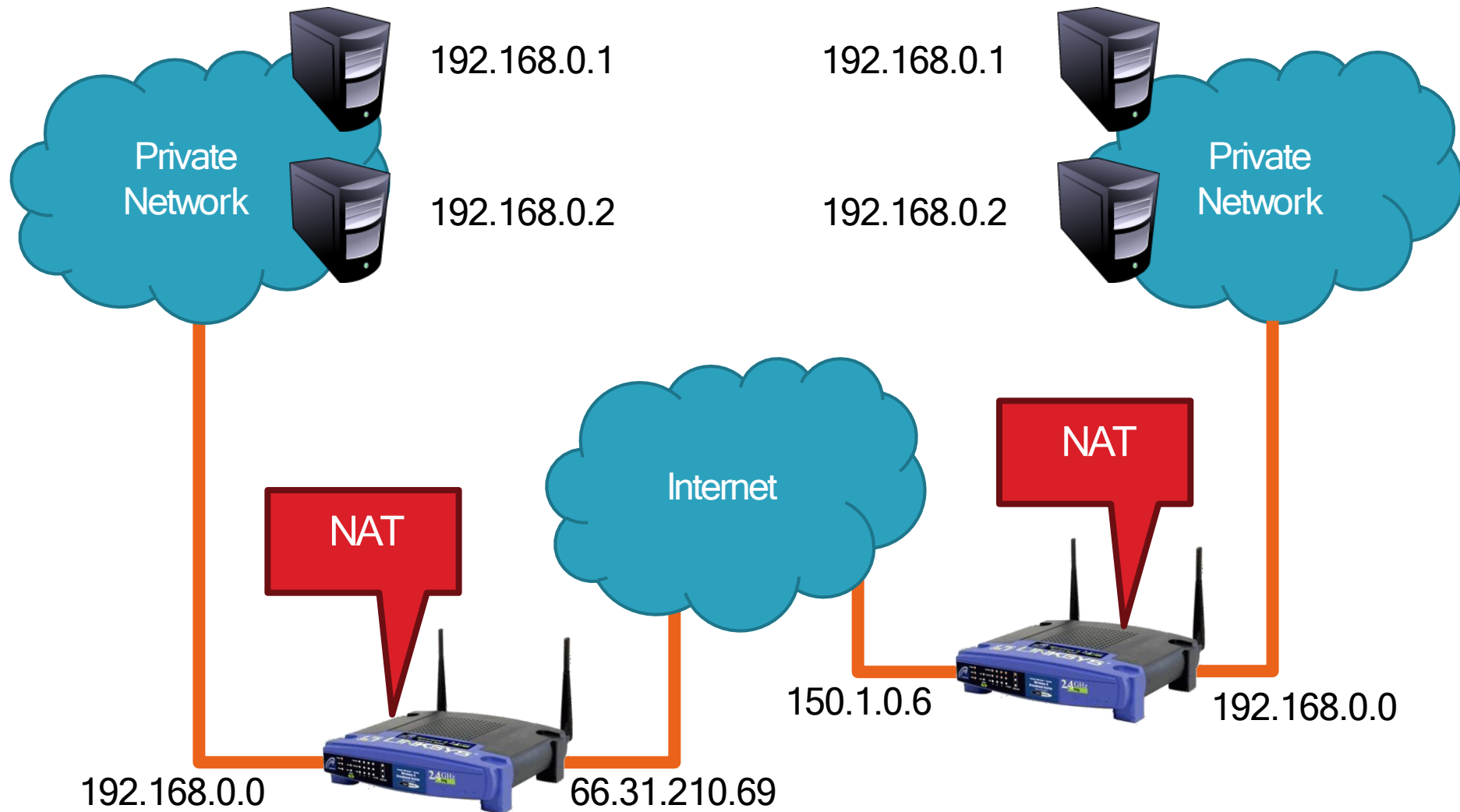
- a) Problem: consumer ISPs typically only give one IP address per-household
 - Additional IPs cost extra
 - More IPs may not be available
- b) Today's households have more networked devices than ever
 - Laptops and desktops
 - TV, bluray players, game consoles
 - Tablets, smartphones, eReaders
- c) How to get all these devices online?

Private IP Networks

- a) Idea: create a range of private IPs that are separate from the rest of the network
 - Use the private IPs for internal routing
 - Use a special router to bridge the LAN and the WAN
- b) Properties of private IPs
 - Not globally unique
 - Usually taken from non-routable IP ranges (why?)
- c) Typical private IP ranges
 - 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255

Private Networks

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Network Address Translation (NAT)

- a) NAT allows hosts on a private network to communicate with the Internet
 - Warning: connectivity is not seamless
- b) Special router at the boundary of a private network
 - Replaces internal IPs with external IP
 - This is “Network Address Translation”
 - May also replace TCP/UDP port numbers
- c) Maintains a table of active flows
 - Outgoing packets initialize a table entry
 - Incoming packets are rewritten based on the table

Basic NAT Operation

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Private Network

Internet

Source: 192.168.0.1
Dest: 74.125.228.67

Source: 66.31.210.69
Dest: 74.125.228.67

Private Address

Public Address

192.168.0.1:2345

66.31.210.69:80



192.168.0.1



66.31.2



74.125.228.67

1

Source: 74.125.228.67
Dest: 66.31.210.69

Source: 74.125.228.67

Dest: 192.168.0.1

Advantages of NATs

- a) Allow multiple hosts to share a single public IP
- b) Allow migration between ISPs
 - Even if the public IP address changes, you don't need to reconfigure the machines on the LAN
- c) Load balancing
 - Forward traffic from a single public IP to multiple private hosts



Natural Firewall

110

Private Network

Internet

Private Address

Public Address



192.168.0.1



66.31.210.69



74.125.228.67

Source: 74.125.228.67: 8679
Dest: 66.31.210.69: 7000

Port Forwarding

112

Private Network

Internet

Private Address

Public Address

192.168.0.1:7000

* * * *
* * * *



192.168.0.1



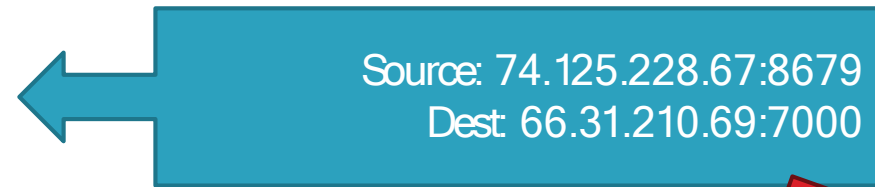
0.69



74.125.228.67

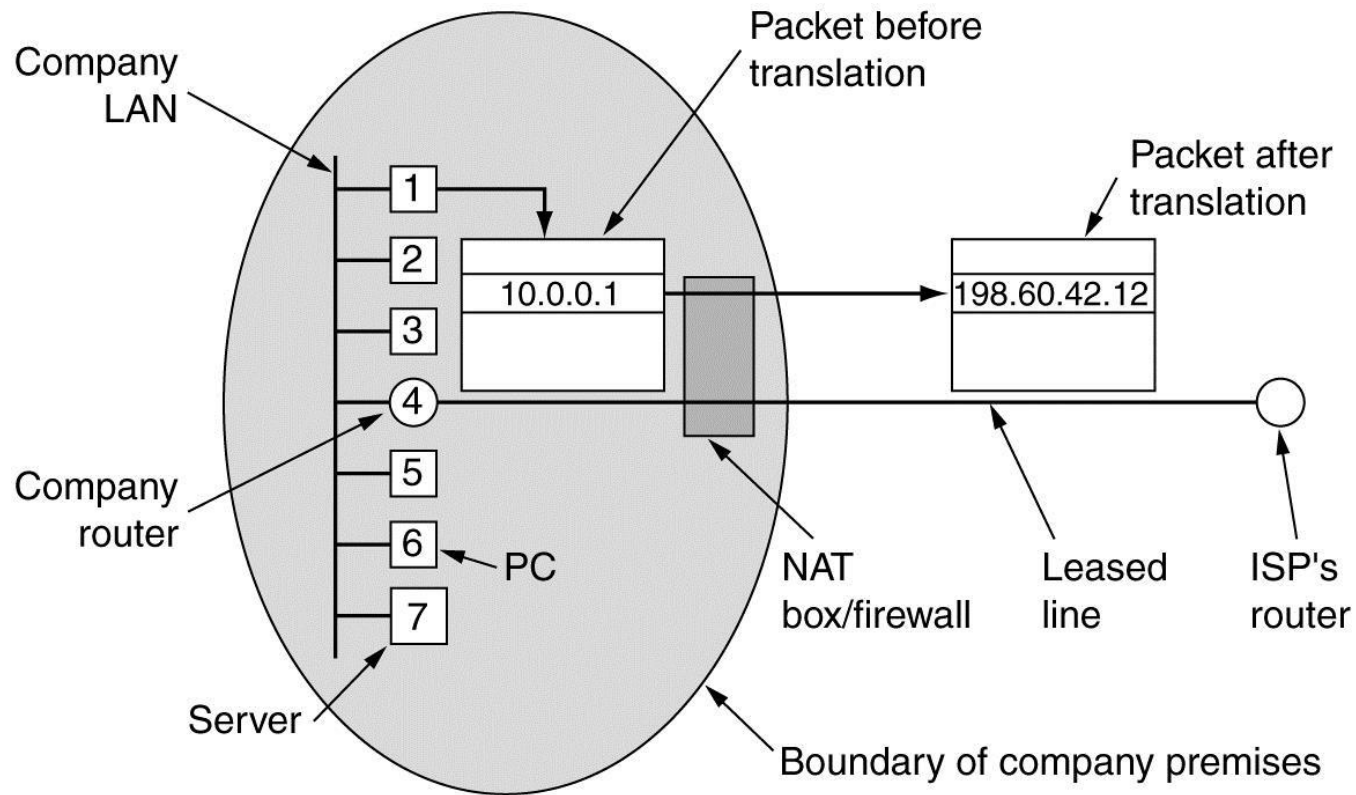


Source: 74.125.228.67:8679
Dest: 192.168.0.1:7000



Source: 74.125.228.67:8679
Dest: 66.31.210.69:7000

NAT – Network Address Translation

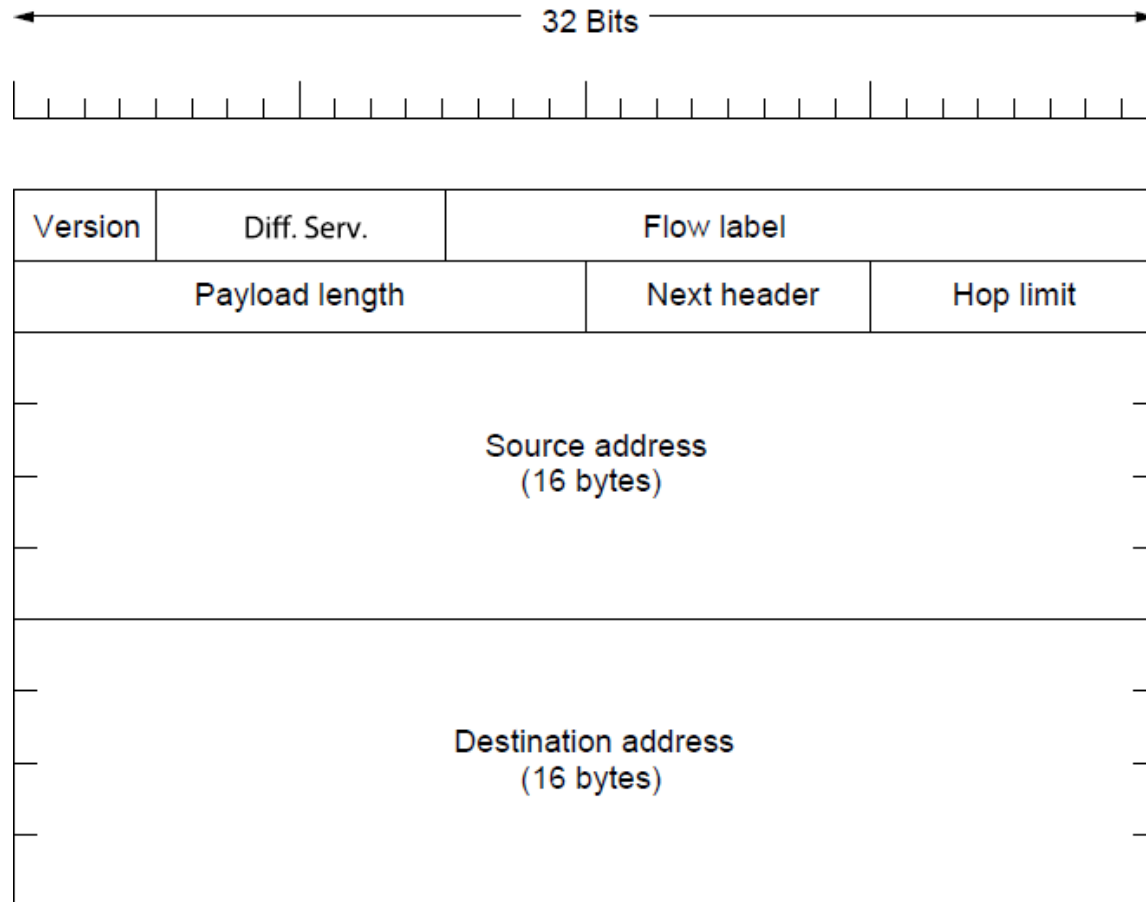


Placement and operation of a NAT box.

IP Version 6 Goals

- Support billions of hosts
- Reduce routing table size
- Simplify protocol
- Better security
- Attention to type of service
- Aid multicasting
- Roaming host without changing address
- Allow future protocol evolution
- Permit coexistence of old, new protocols. . .

IP Version 6 (1)



The IPv6 fixed header (required).

IP Version 6 (2)

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

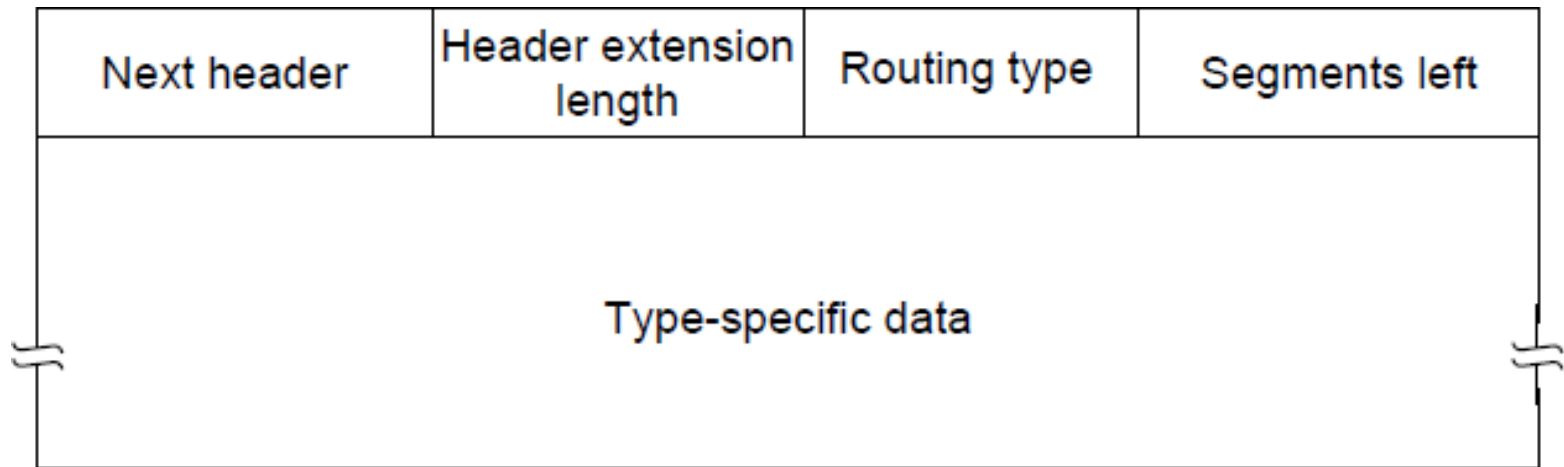
IPv6 extension headers

IP Version 6 (3)

Next header	0	194	4
Jumbo payload length			

The hop-by-hop extension header for large datagrams (jumbograms).

IP Version 6 (4)



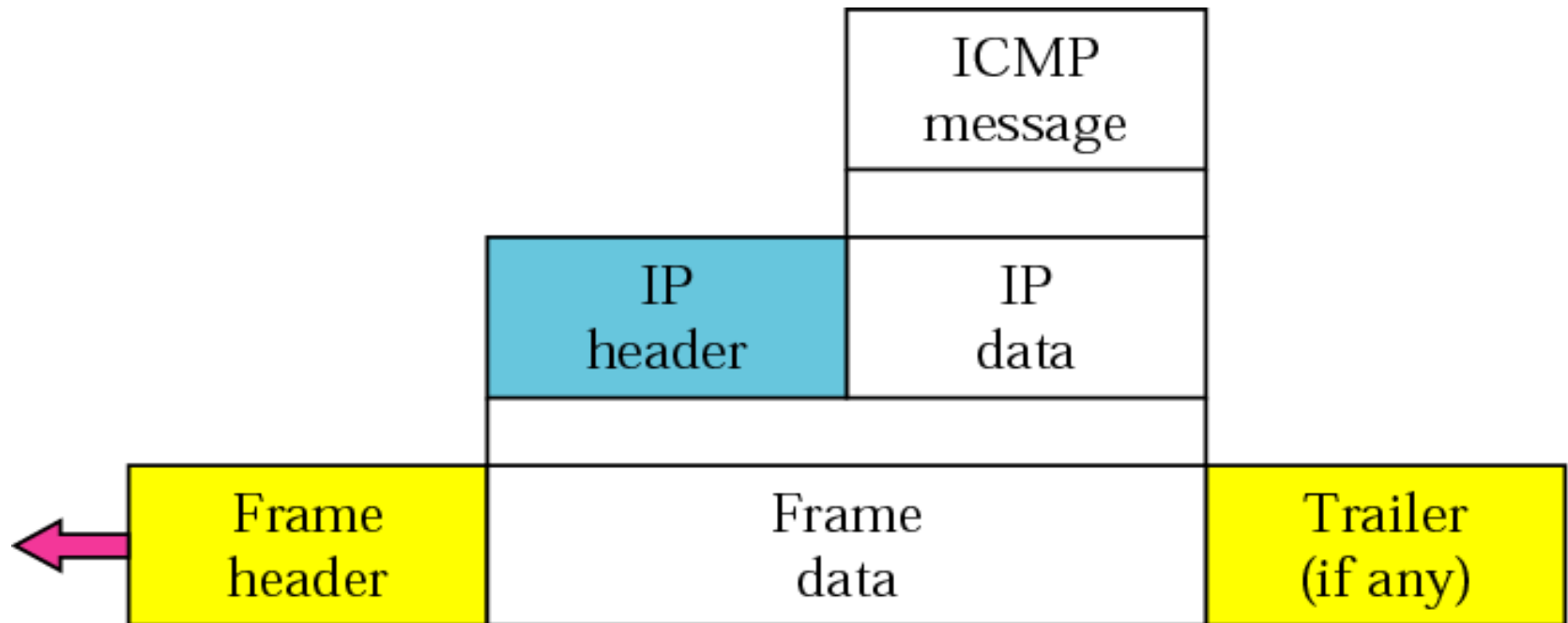
The extension header for routing.

Internet Control Protocols (1)

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and Echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

The principal ICMP message types.

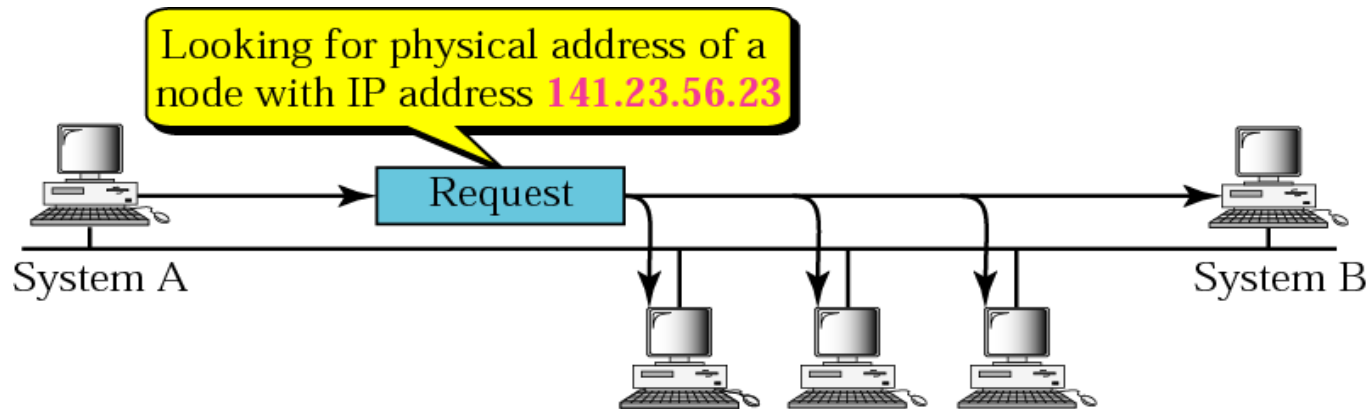
ICMP encapsulation



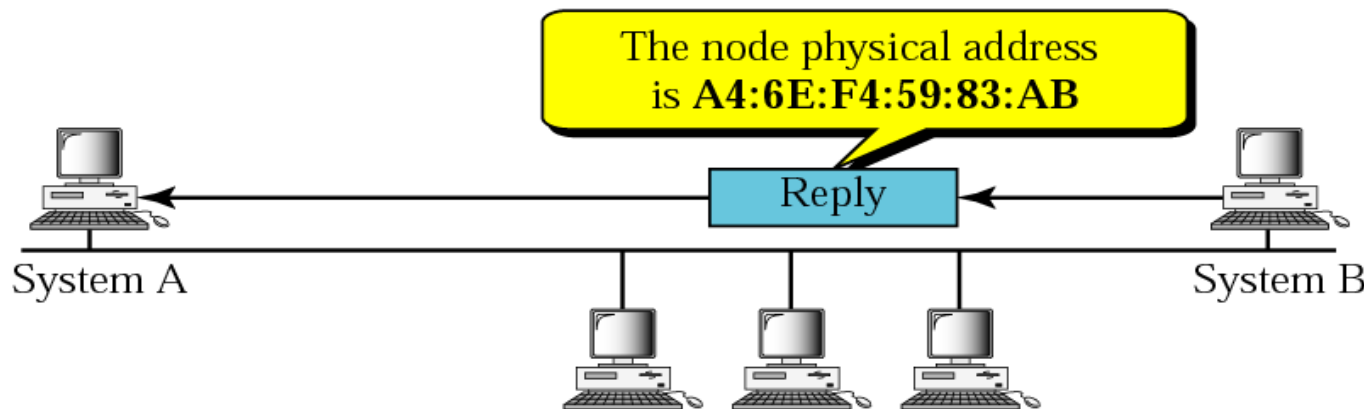
ICMP messages

<i>Category</i>	<i>Type</i>	<i>Message</i>
Error-reporting messages	3	Destination unreachable
	4	Source quench
	11	Time exceeded
	12	Parameter problem
	5	Redirection
Query messages	8 or 0	Echo request or reply
	13 or 14	Timestamp request or reply
	17 or 18	Address mask request or reply
	10 or 9	Router solicitation or advertisement

ARP– The Address Resolution Protocol



a. ARP request is broadcast

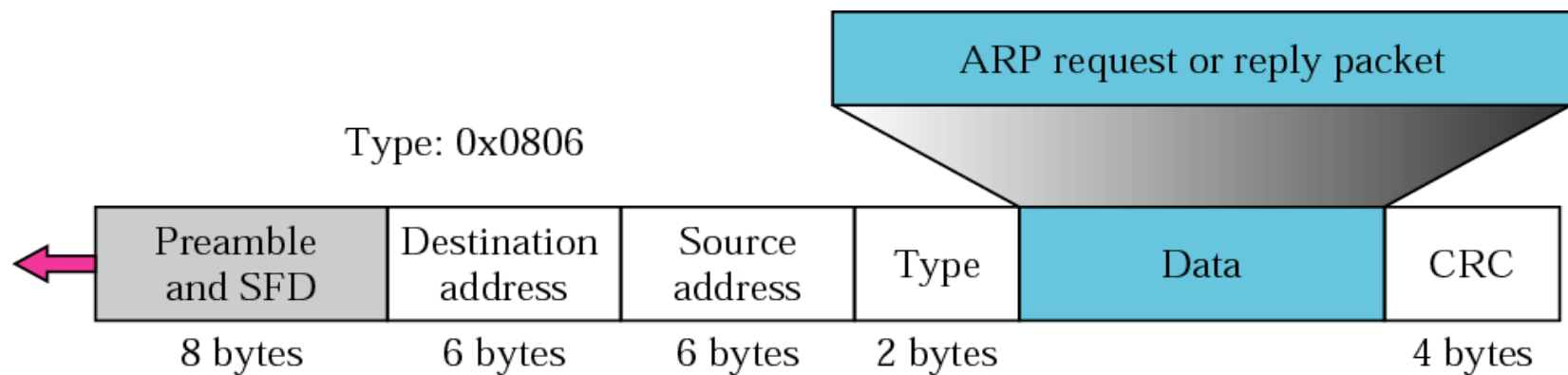


b. ARP reply is unicast

ARP packet

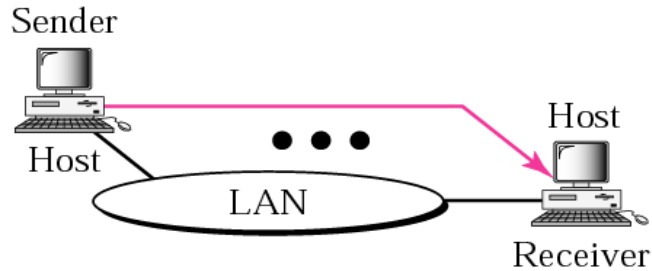
Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Encapsulation of ARP packet



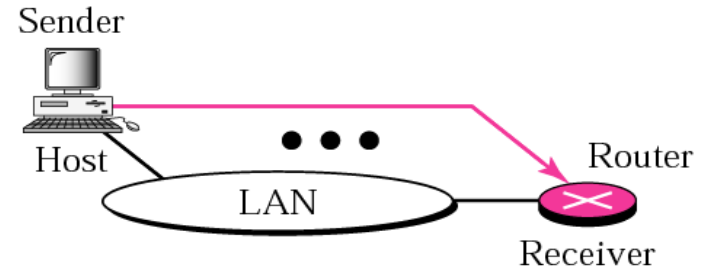
Four cases using ARP

Target IP address:
Destination address in the IP datagram



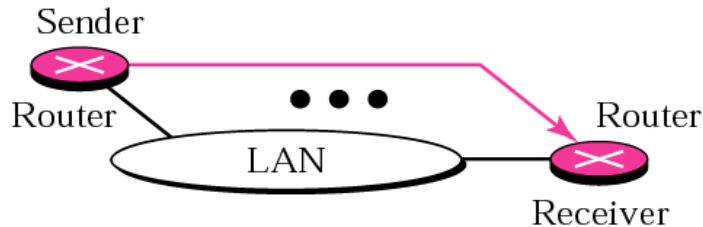
Case 1. A host has a packet to send to another host on the same network.

Target IP address:
IP address of a router



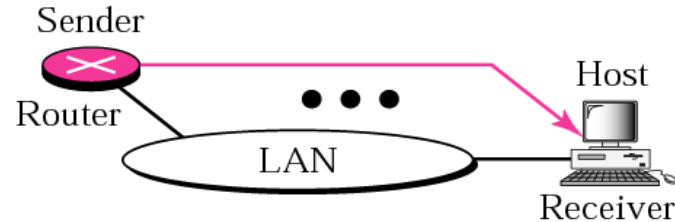
Case 2. A host wants to send a packet to another host on another network. It must first be delivered to a router.

Target IP address:
IP address of the appropriate router
found in the routing table



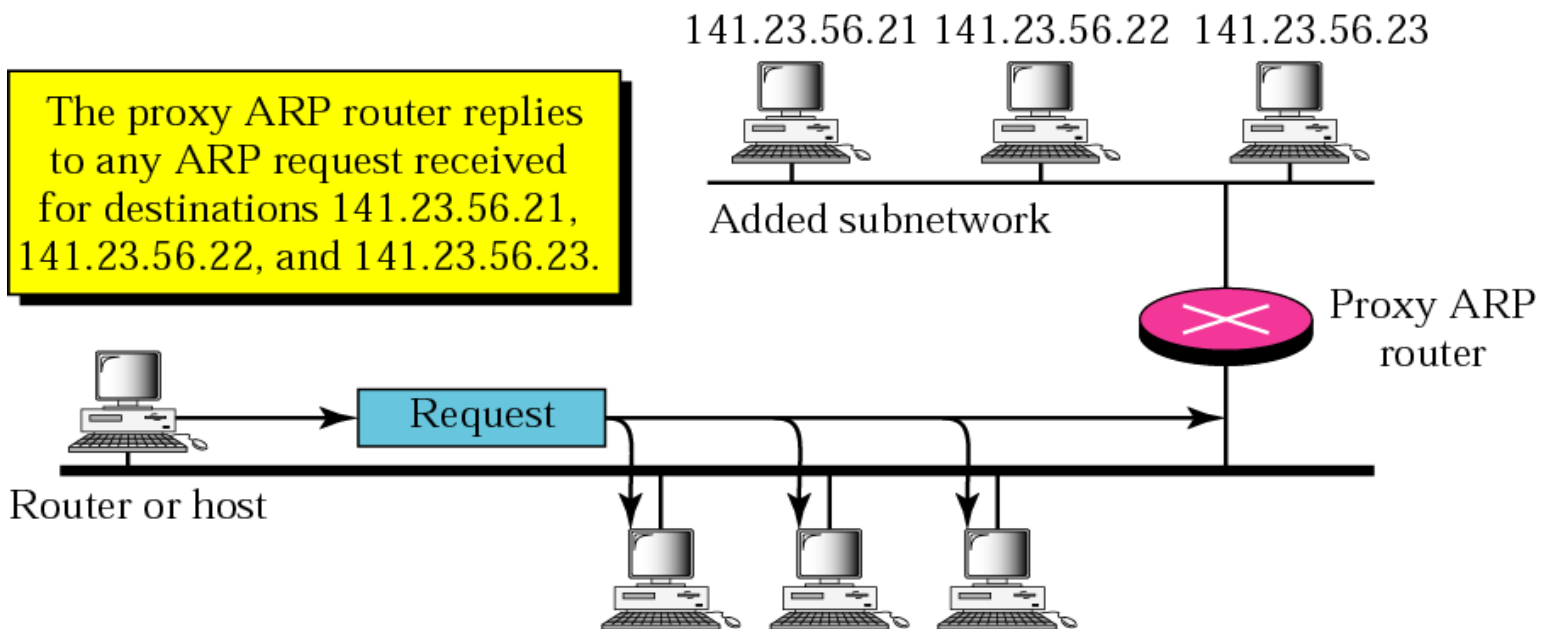
Case 3. A router receives a packet to be sent to a host on another network. It must first be delivered to the appropriate router.

Target IP address:
Destination address in the IP datagram

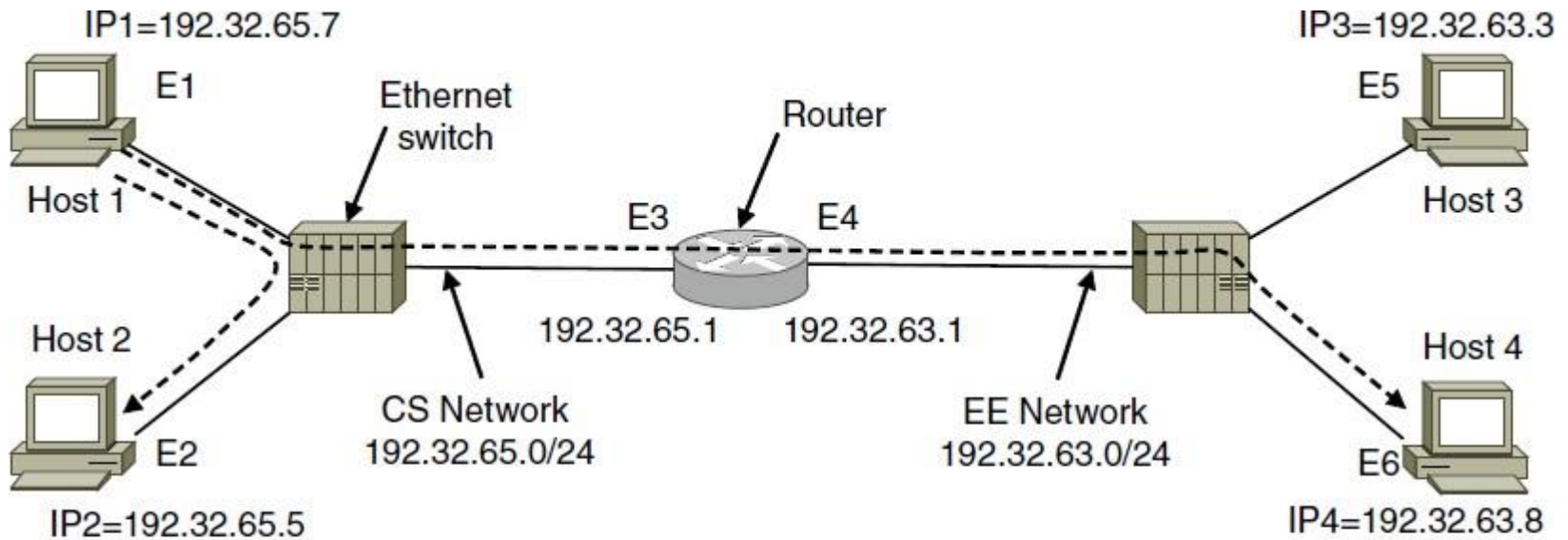


Case 4. A router receives a packet to be sent to a host on the same network.

Proxy ARP



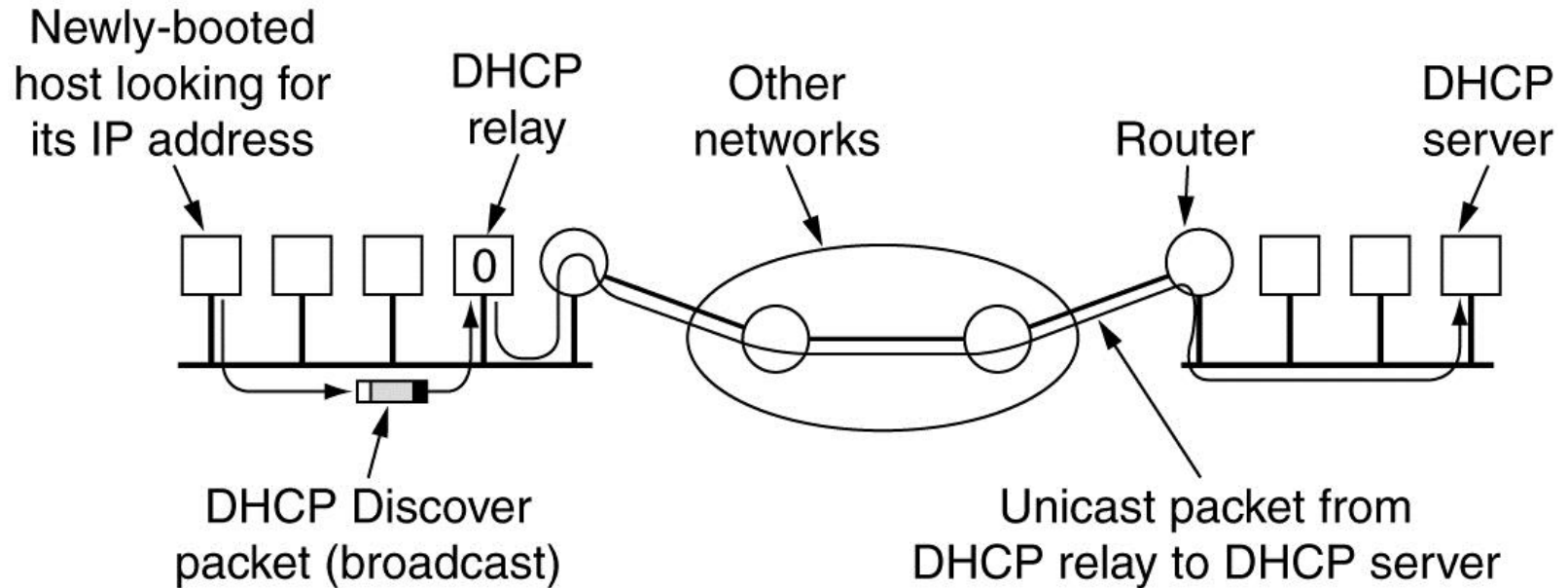
ARP



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

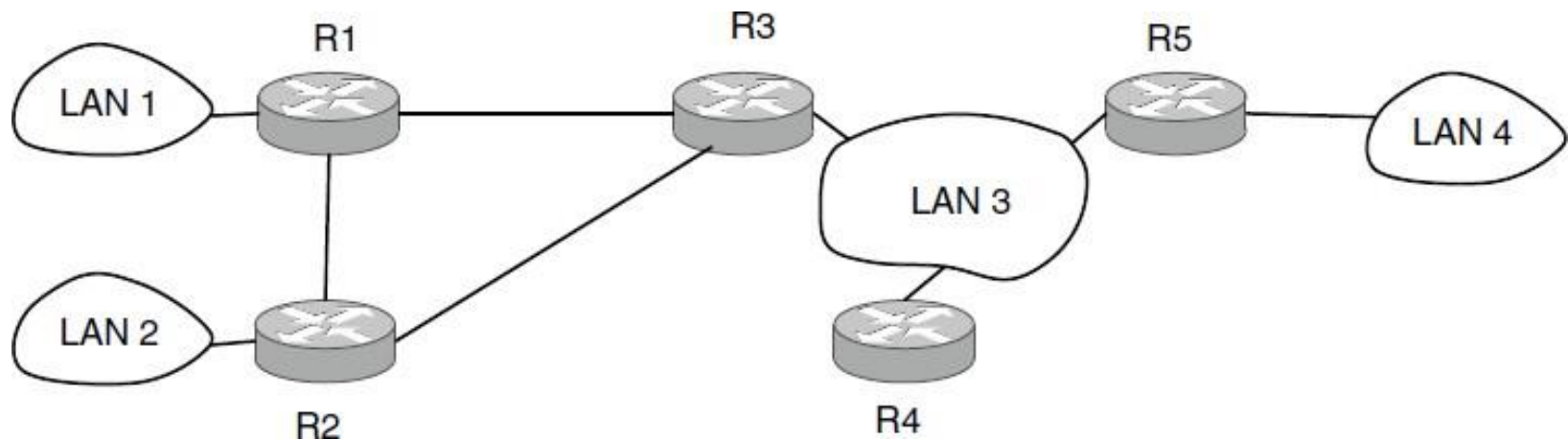
Two switched Ethernet LANs joined by a router

Dynamic Host Configuration Protocol



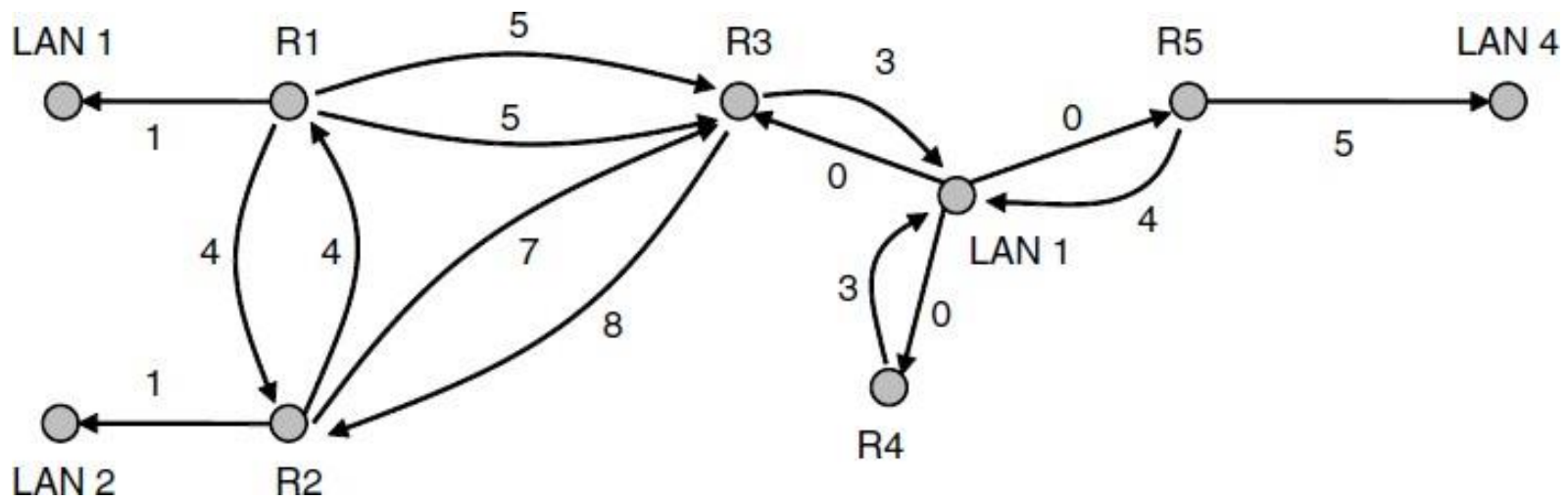
Operation of DHCP.

OSPF—An Interior Gateway Routing Protocol (1)



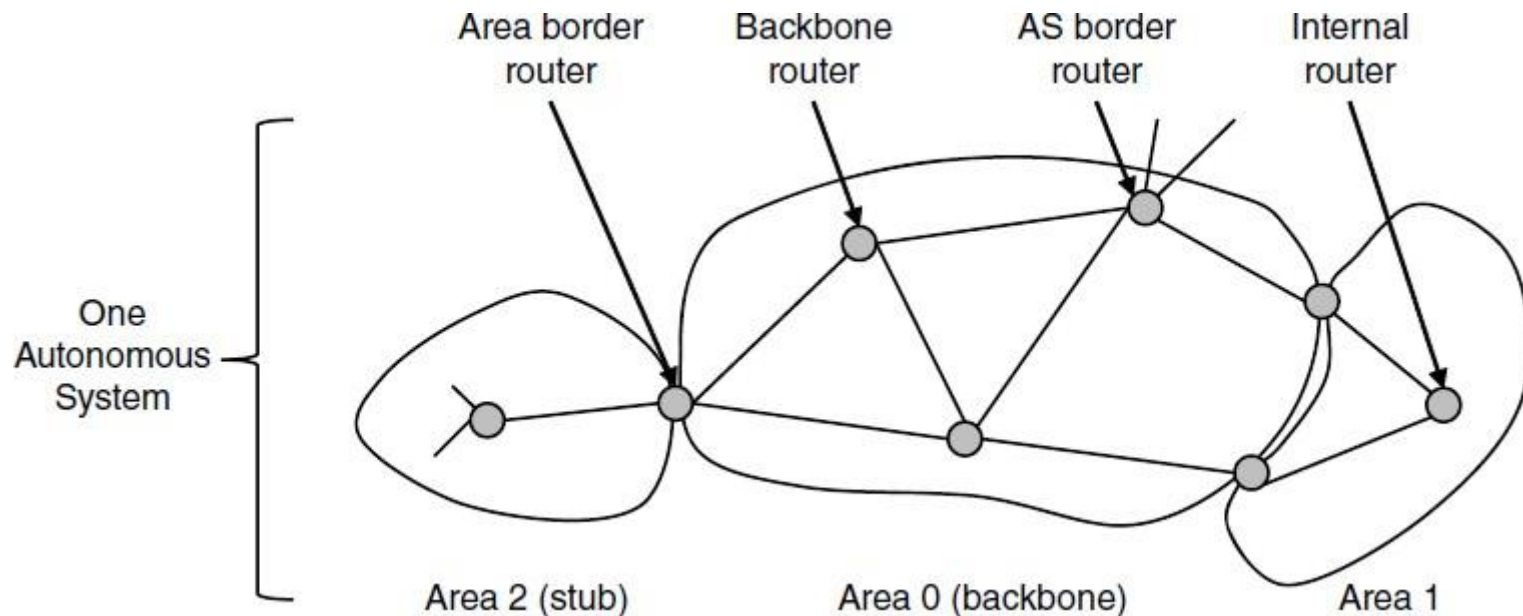
An autonomous system

OSPF—An Interior Gateway Routing Protocol (2)



A graph representation of the previous slide.

OSPF—An Interior Gateway Routing Protocol (3)



The relation between ASes, backbones, and areas in OSPF.

OSPF—An Interior Gateway Routing Protocol (4)

Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

The five types of OSPF messages

BGP—The Exterior Gateway Routing Protocol (1)

Examples of routing constraints:

- No commercial traffic for educat. network
- Never put Iraq on route starting at Pentagon
- Choose cheaper network
- Choose better performing network
- Don't go from Apple to Google to Apple

End

Chapter 5